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Financialization and the non-financial corporation: An investigation of firm-level investment behavior in the United States

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Abstract

Changes in the portfolio and financing behavior of non-financial corporations (NFCs) over the post-1970 period in the U.S. economy point to the financialization of the NFC and raise the question of accompanying changes in fixed investment behavior. Using a firm-level panel, this article econometrically investigates the relationship between financialization and investment, exploring the implications of changes in financing behavior, increasingly entrenched shareholder value norms and rising firm-level demand volatility for NFC investment in the U.S. economy between 1971 and 2013. Shareholder value orientation is, in particular, identified as a characteristic of the post-1970 U.S. economy associated with a statistically and economically significant decline in NFC investment rates. The stock of financial assets, conversely, is found to be a positive correlate of firm investment. The analysis also highlights key differences by firm size. In particular, shareholder value norms are found to primarily influence the investment behavior of large NFCs, while rising volatility most substantially impacts small firms.

1 | INTRODUCTION

The increasingly dominant role of finance over the post-1970 period in the U.S. has, in recent years, led to a growing literature on financialization. While the precise concept of financialization varies considerably across analyses, the shared premise is that financial sector growth signifies an important structural change in the U.S. economy, highlighted by sustained growth in the share of financial profits in total corporate profits (Krippner, 2012). With respect to non-financial business, financialization is manifested in an increasingly complex relationship between non-financial corporations (NFCs) and the financial sector. As many large NFCs increasingly resemble financial companies, the hostile takeover movement and the emergence of shareholder value ideology point to changes in corporate governance

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arguably increasing the weight of short-term valuations of firm performance in managerial decision-making (Crotty, 2005). Changes in NFC financial behavior are reflected in an increasing share of financial assets in firm portfolios, as well as in changes in the structure of external finance, including increasing indebtedness and growing equity repurchases among large firms.

This article analyzes changes in fixed investment rates in the post-1970 U.S. economy via an econometric investigation of firm-level investment behavior. Recent empirical work provides support for the contention that changes associated with a broadly-defined phenomenon of financialization inhibit fixed investment. At the aggregate level, Stockhammer (2004) finds that rising rentiers' income explains roughly one third of a slowdown in capital accumulation in the United States (p. 736), and van Treeck (2008) argues that rising rentier incomes are responsible for a diversion of funds from physical investment into consumption expenditure. At the firm level, Orhangazi (2008) argues that increased payments by NFCs to finance and higher financial profits earned by NFCs constrain fixed investment, particularly among large firms. In an emerging market context, Demir (2009) similarly finds that financial liberalization—by increasing both uncertainty and the rate of return gap between financial assets and fixed capital—reduces fixed investment, while supporting the acquisition of financial assets.

Orhangazi (2008) is, in particular, often cited as firm-level evidence that 'financialization' in the U.S. economy depresses fixed investment. However, while Orhangazi's paper points to important correlations between investment and flows of income between NFCs and the financial sector (financial payments and financial profits), it also raises further questions. In particular, why has firm behavior changed specifically over the post-1970 period, such that these indicators of financialization grow in a sustained way? The discussion suggests possible channels, including shareholder value orientation and increasingly short-term managerial time horizons; however, links between these possible mechanisms and the empirical measures of financialization that are employed are unclear. Importantly, financial profits earned by NFCs and financial payments made by NFCs are the result of firm decisions to acquire financial assets, or to borrow, repurchase stock, or pay dividends. The implications for fixed investment—and, thus, conclusions about financialization and investment—vary with the motivation driving changes in NFC behavior.

Take, for example, Orhangazi's finding that increased payments from NFCs to finance reduce fixed investment. Financial payments—reflecting, for example, higher interest payments—draw (by definition) on a firm's pool of available funds and, therefore, come at a short-run tradeoff with other uses of funds, including physical investment. Conversely, higher leverage—and, correspondingly, higher interest payments—is the result of a firm's decision to borrow in pursuit of some objective, whether profits, growth, a stock price increase, or to cover rising interest obligations. The implications for fixed investment vary with this objective: higher financial payments that stem from borrowing to finance (profitable) fixed investment differ from borrowing to buyback stock.

The question, consequently, arises of why NFC leverage grew over the post-1970 period and, more generally: what factors underlie the observed changes in NFC financial behavior, and with what implications for fixed investment? This article explores this question via an econometric investigation of a firm-level investment function, in which the empirical specification introduces increasingly entrenched shareholder value norms as a key change in the context within which NFCs make investment and financing decisions that is *specific* to the post-1970 U.S. economy. The empirical results highlight the statistical and economic significance of growing shareholder value orientation as an important factor inhibiting fixed investment in the post-1970 U.S. economy. These results indicate that, while changes in investment behavior are linked to financial decisions, they are *rooted* in new corporate governance norms.

This evidence regarding shareholder value orientation corroborates the branch of the financialization literature emphasizing the adverse implications of shareholder value orientation for key non-financial outcomes including investment, sustainable prosperity and employment (Lazonick, 2009a; Lazonick & O'Sullivan, 2000). This article is the first, however, to provide direct econometric evidence that

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shareholder value orientation is one important mechanism depressing firm-level investment rates in the post-1970 U.S. economy. Note that, while Orhangazi (2008) is often cited as evidence to this effect, the key empirical indicator—financial payments—includes payments to both creditors and shareholders, such that shareholder payouts are not explored independently. The analysis presented here, in turn, analyzes changes in corporate governance norms associated with shareholder value orientation in a novel way: shareholder value orientation is proxied via average yearly industry-level repurchases, to investigate the impact on investment rates of increasingly-entrenched norms that encourage managers to target stock market-based indicators of firm performance when making investment and financing decisions.

In addition to the results regarding shareholder orientation, the empirical results indicate that certain changes in NFC financial structure—most notably, increased holdings of financial assets—are positively correlated with fixed investment. Accordingly, the analysis indicates that different dimensions of financialization have counteracting impacts on investment behavior, and points to the importance of isolating relationships between specific features of financialization and investment behavior. Furthermore, the empirical findings capture a negative effect of firm-level volatility on investment; this mechanism is important to note given significant growth in firm-level volatility concurrent with the financialization of the U.S. economy, and also that volatility is likely linked to the acquisition of financial assets. Finally, this article emphasizes systematic behavioral differences by firm size: while shareholder value norms significantly impact large firms' investment rates, smaller firms' investment is found to be more strongly inhibited by rising volatility.

The empirical specification in this article draws on theories of investment in the spirit of Keynes and Minsky, which impart a key role to financial factors in investment decisions. Because the decision to invest involves not only a decision about the proposed investment, but also a decision about how to finance that investment, a firm's investment and financing decisions are interdependent. A large empirical literature based on this body of theory confirms the relevance of financial factors in explaining investment behavior (see, e.g., Brown, Fazzari, & Petersen, 2009; Fazzari & Mott, 1986; Kuh & Meyer, 1957; Ndikumana, 1999). Finally, the econometric analysis, as well as the descriptive discussion in Section 2, utilizes firm-level data, thereby incorporating important firm-level variation and heterogeneity. It is relevant to note, however, that additional analysis would be necessary to draw direct conclusions about aggregate accumulation from the firm-level analysis, given possible interactions between micro behavior and macro outcomes. Changes in firm-level investment influence goods and financial market equilibrium and may, for example, induce demand effects such that aggregate spending falls (Skott & Ryoo, 2008, p. 834).

The article is organized as follows: Section 2 summarizes changes in NFC financial structure from 1971 to 2013; Section 3 outlines a framework motivating the econometric specification; Section 4 lays out the empirical specification and data; Section 5 presents the empirical results; and Section 6 concludes.

2 | THE 'FINANCIALIZATION' OF THE NFC

2.1 | NFC financial decisions: Portfolio composition and external finance

Trends in the structure of firm-level balance sheets summarize changes in NFC portfolio and financing behavior in the post-1970 U.S. economy that point to the 'financialization' of non-financial firms.[1] On the asset side of firm balance sheets there is a marked decline in the share of fixed capital held in NFC portfolios: between 1971 and 2013 the across-firm yearly median of fixed capital relative to sales, shown by the black line in Figure 1a, declines 7.1% points, from 24.3% in 1971 to 17.2% in 2013.[2]

[1]The data in this article is from Standard & Poor's Compustat annual industrial database. Variables are defined in Section 4.4 and summarized in the Appendix.

[2]The qualitative trend holds when measured relative to total assets. Sales are used here to proxy for firm size.

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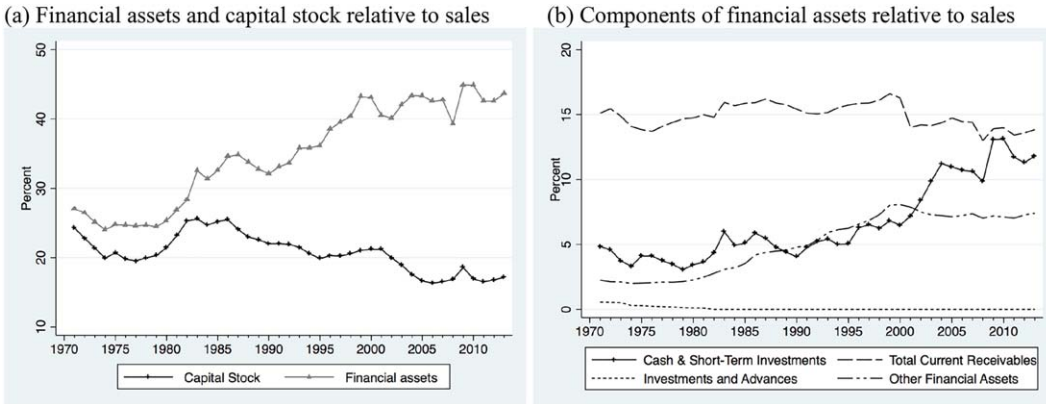

FIGURE 1 Financial assets and capital relative to sales, 1971–2013 [Colour figure can be viewed at wileyonlinelibrary.com]

Note. These figures plot the across-firm yearly medians of each asset component measured relative to sales. See details on variable definitions in Table A1.

Source. Compustat, author's calculations

Concurrently, financial assets relative to sales rise 16.6% points, from 27.0% in 1971 to 43.7% in 2013. This portfolio shift toward financial assets has been cited in the financialization literature to motivate a possible 'crowding out' relationship between financialization and fixed investment, whereby financial investments increasingly replace investment in physical assets.

Growth in NFC financial assets holdings is concentrated, first, in liquid short-term investments and, second, in 'other' financial assets. Figure 1b decomposes total financial assets into four (exhaustive) subcategories: total current receivables, cash and short-term investments, investments and advances and 'other' financial assets.[3] The first panel of Table 1 summarizes the change in total financial assets, each subcategory and the capital stock between 1971 and 2013. During this period, current receivables and advances both grow relatively proportionally to firm-level sales. Simultaneously, 'other' financial assets rise from 2.3% of sales in 1971 to 7.4% in 2013. The largest increase, however, is in liquid financial assets, which rise from 4.8% of sales in 1971 to 11.8% in 2013.[4] Importantly, these trends are not the consequence of changes in the industrial composition of the sample and, instead, hold across industrial sub-samples, including industries associated with important structural changes over this period (including tech, manufacturing and services) (Davis, 2016, p. 126).

Furthermore, this portfolio shift toward financial assets occurs across firm size.[5] Accordingly, the median portfolio shift in Figure 1 is not simply driven by a large number of very small firms with little aggregate importance. The second and third panels of Table 1 summarize the change in each portfolio component between 1971 and 2013 for sub-groups of small and large firms, where small firms are defined as firms with total assets in the bottom quartile of the asset distribution in a given year, and large firms as firms with total assets in the top quartile of the asset distribution. Among small firms, total financial assets rise from 28.6% of sales in 1971 to 50.0% in 2013, while fixed capital declines from 17.5% of sales to 7.1%. Similarly, total financial assets held by large firms increase from 28.8% of sales in 1971 to 42.4% in 2013, and fixed capital declines from 49.3 to 30.7% of sales. As with the

[3]Unfortunately, the documentation describing 'other' financial assets is unilluminating, and a similar issue arises in the aggregate data. For further discussion, see Crotty (2005) and Davis (2016).

[4]Because of accounting rules, cash cannot be disaggregated from other short-term assets, which includes securities with original maturities less than 1 year.

[5]As such, the trend in across-firm yearly *mean* portfolio composition is analogous to the trend in the yearly median.

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TABLE 1 Changes in components of financial assets relative to sales for small and large firms; medians

	All firms			Small firms			Large firms		
	1971	2013	pp change	1971	2013	pp change	1971	2013	pp change
Total financial assets	27.01	43.65	16.64	28.60	50.00	21.40	28.78	42.40	13.62
Cash & Short-term assets	4.84	11.81	6.97	5.65	19.20	13.55	4.98	7.80	2.82
Receivables	15.12	13.84	−1.28	16.15	12.05	−4.10	14.56	13.76	−0.81
Advances	0.57	0.00	−0.57	0.00	0.00	0.00	2.97	1.25	−1.72
'Other' Financial Assets	2.25	7.39	5.14	2.26	5.56	3.30	2.49	9.74	7.25
Capital	24.27	17.16	−7.11	17.52	7.13	−10.39	49.28	30.73	−18.55

Note. See details on variable definitions in Table A1. A firm is categorized as small if its total assets are in the bottom quartile of the asset distribution for any given year. A firm is categorized as large if its total assets are in the top quartile of the asset distribution for any given year. The qualitative trends are robust to alternative definitions of firm size. Note that cash cannot be disaggregated from other short-term securities.
Source. Compustat, author's calculations.

full sample, growth in financial asset holdings is concentrated in 'cash' and in 'other' financial assets for both small and large firms. Small firms, however, have acquired relatively greater shares of liquid assets, whereas large firms' financial asset acquisitions are concentrated in 'other' financial assets. Among small firms, liquid financial assets rise from 5.7% of sales in 1971 to 19.2% in 2013, while 'other' financial assets rise from 2.3 to 5.7% of sales. Among large firms, conversely, financial asset acquisitions are dominated less by liquid assets, whereas 'other' financial assets increased from 2.5 to 9.7% of sales from 1971 to 2013.

While the shift in NFC portfolio composition occurs across the distribution of firms, albeit to varying degrees, changes in the structure of both debt and equity differ decisively by firm size. An increase in gross corporate debt has been cited as a definitive characteristic of the financialization of NFCs (Palley, 2007), and Flow of Funds data clearly documents rising leverage at the sector-level. Firm-level data, however, highlights that rising mean leverage across NFCs—defined as debt relative to the capital stock—occurs simultaneous with a small *decline* in median leverage shown in Figure 2.[6] Furthermore, since the early 1970s, the distribution of debt among small firms is increasingly skewed toward zero, whereas among large firms, the distribution of debt shifts to the right, such that fewer large firms have 'low' leverage in 2013 than in the early 1970s. Together, these trends highlight growing leverage among large firms and concurrent deleveraging among smaller firms. Furthermore, these firm-size differences are again not dominated by changes in the industrial composition of the non-financial corporate sector and, in contrast, to firm-size differences, each industry closely follows the full sample patterns (see Davis, 2016 for additional details).

Rising debt among large firms is accompanied by a dramatic increase in repurchases of own stock. Stock repurchases have received considerable attention in reference to the shareholder value movement (Lazonick, 2009a; Lazonick & O'Sullivan, 2000), and sector-level growth in buybacks is well known. The firm-level data reinforces the aggregate trend, but also highlights that repurchases are concentrated among large firms. Figure 3 plots the across-firm yearly mean of equity repurchases relative to outstanding equity for the full sample of firms and by firm size. While repurchases among large firms closely follow the full-sample pattern, they remain low among small firms for the entire period.

[6]Debt relative to total assets declines in both the mean and the median, indicating that the same firms that are more indebted also hold more cash. From 1971 to 2013, mean debt to total assets declines moderately from 33.3 to 30.8%, while the across-firm median declines from 27.0 to 21.5%.

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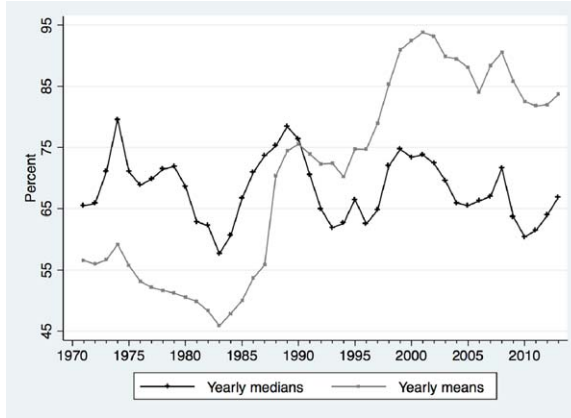

FIGURE 2 Debt relative to the capital stock, 1971–2013 [Colour figure can be viewed at wileyonlinelibrary.com]
Note. Debt is defined as total debt (the sum of both current and long-term debt). This figure plots the across-firm yearly medians and across-firm yearly means of debt relative to the stock of physical capital. See additional details on variable definitions in Table A1.
Source. Compustat, author's calculations

Furthermore, median buybacks in any given year, both for the full sample and each sub-sample of firms, are zero, reflecting the bulky and episodic nature of repurchase plans: firms announce that stock will be repurchased over a set number of years, followed by years without repurchases. Figure 3 also highlights volatility in repurchase growth, which reflects strong procyclicality (Jagannathan, Stephens, & Weisback, 2000), and also correlates with SEC rule changes supporting repurchases (Davis, 2016).

Concurrent with these changes in the structure of external finance, the correlation between new borrowing and investment—shown in Figure 4—declines, indicating that rising leverage among large firms is not channeled into physical investment.[7] Holmstrom and Kaplan (2001) argue, for example, that leveraged buyouts during the hostile takeover movement contributed to rising corporate debt during the 1980s. Borrowing to buyout a company has no direct link to capital investment. The same is true of repurchasing stock. The concurrent rise in debt and repurchases over this period, therefore, suggests that equity is increasingly replaced by debt on large firms' balance sheets, while 'traditional' financing behavior—debt finance for the acquisition of physical capital—breaks down. As with changes on the asset side of NFC balance sheets, changes in the structure of external finance, therefore, raise questions about fixed investment in the post-1970 U.S. economy.

These changes in the structure of NFC balance sheets have occurred simultaneous with a secular rise in firm-level volatility. Rising firm-level volatility has been extensively documented in the existing literature (see Comin & Phillipon, 2005), and has been linked, for example, to new information and communication technologies leading to shorter product life cycles (Skott & Guy, 2013). Figure 5 plots demand volatility for firms in this sample, defined as the coefficient of variation in the firm-level sales-to-capital ratio, where the standard deviation (*SD*) and mean are based on 5 years of lags. Volatility for the full sample of firms, shown by the solid black line, rises from 18.0 to 29.2% between 1971 and 2013, peaking at 35% in 2004 and 2005. Concurrently, volatility among large firms increases from 11.7% in 1971 to 15.8% in 2013. Compared to large firms, however, volatility among small firms increased far more dramatically, nearly doubling from 29.6 to 59.9% from 1971 to 2013. The trends in volatility for industrial subsamples, conversely, again closely track the full-sample trends. Figure 5, therefore, suggests that rising firm-level demand volatility provides particular insight into the behavior of small firms in the post-1970 U.S. economy.

[7]A previous version of this graph appears in Mason (2013).

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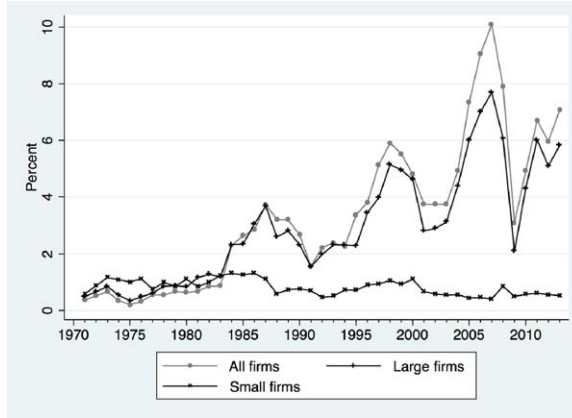

FIGURE 3 Equity buybacks, 1971–2013 [Colour figure can be viewed at wileyonlinelibrary.com]

Note. This figure plots the across-firm yearly means of gross equity repurchases relative to total outstanding equity. See additional details on variable definitions in Table A1. A firm is categorized as small if its total assets are in the bottom quartile of the asset distribution for any given year. A firm is categorized as large if its total assets are in the top quartile of the asset distribution for any given year.

Source. Compustat, author's calculations

In particular, higher volatility, reflecting greater uncertainty, is likely to drive higher demand for liquid assets. Bates, Kahle, and Stulz (2009) find that idiosyncratic risk (firm-level volatility) is a determinant of increased cash holdings among U.S. firms over this period (p. 2018).[8] This evidence is consistent with the fact that small firms hold relatively greater shares of liquid—as opposed to non-liquid—financial assets than large firms (see Table 1). Higher volatility and correspondingly greater uncertainty regarding future demand may similarly be a factor behind small firms' de-leveraging. In turn, volatility may also influence the decision to invest in fixed capital. Capital investments are long-term and largely irreversible, such that in a more volatile environment, investment demand is likely to be lower for given expected returns. Along these lines, in an emerging market context, Demir (2009) emphasizes the role of increased uncertainty in driving a portfolio reallocation away from long-term irreversible fixed capital, and toward (reversible) financial assets.

2.2 | Changing corporate governance norms: Shareholder value ideology

Growing shareholder value orientation among NFC managers is one factor that has likely shaped the post-1970 changes in firm behavior that point to the financialization of non-financial firms. Institutional changes supporting shareholder value principles began in the 1970s—as inflation increased the value of corporate plant and equipment relative to low stock prices, supporting the emergence of a corporate takeover market (Krippner, 2012)—and became increasingly entrenched over the 1980s and 1990s, with the rise of institutional investors, changing norms regarding managerial pay, and agency theory (Lazonick & O'Sullivan, 2000). Institutional investors, for example, have supported a transition from long-term stock holding toward higher trading frequency (Lazonick & O'Sullivan, 2000; Stout, 2012), as well as a push for stock-option based managerial compensation (Lazonick, 2009b). Concurrently, agency theory suggested mechanisms designed to alleviate agency problems between managers and shareholders (owners) within firms, purportedly improving firm performance. These mechanisms include a hostile market for corporate control that 'disciplines' managers via a threat to managerial

[8]An alternative explanation for the cash build-up among NFCs emphasizes tax motives: because profits earned abroad would be taxed if repatriated, firms hold foreign profits as cash (Foley et al., 2007). This explanation would primarily apply to large NFCs, which are more likely to earn foreign income. Bates et al. (2009) find, however, that even firms without foreign income exhibit a secular increase in cash holdings.

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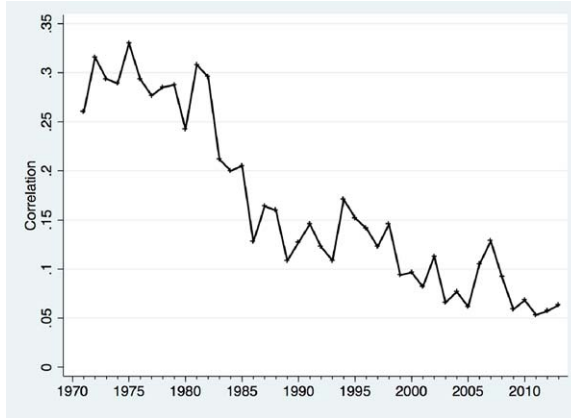

FIGURE 4 Correlation between investment and borrowing over time, 1971–2013 [Colour figure can be viewed at wileyonlinelibrary.com]

Note. Borrowing is defined as the year-on-year change in total outstanding debt. See additional details on variable definitions in Table A1.
Source. Compustat, author's calculations

autonomy (Jensen, 1986, p. 324), and stock-option based executive compensation (Jensen & Murphy, 1990).

These changes have gradually led to the internalization of 'value maximization' as a motive driving managerial decision-making. This emphasis on 'value maximizing' objectives is summarized by an introductory corporate finance text, which defines the 'fundamental objective of corporate finance: maximizing the current market value of the firm's outstanding shares. . .[The] objective overrides other plausible goals, such as "maximizing profits"' (Brealey & Myers, 2012, p. 13). Observed changes in firm-level balance sheets, furthermore, suggest that this shift in objectives has influenced managers' portfolio and financing decisions. Stock buybacks, in particular, are a clear manifestation of shareholder value-maximizing objectives: buybacks improve (stock) market-based valuations of firm performance, reflected in both a higher share price and higher return on equity. Thus, buybacks both diminish the likelihood of hostile takeover and increase the value of stock options.

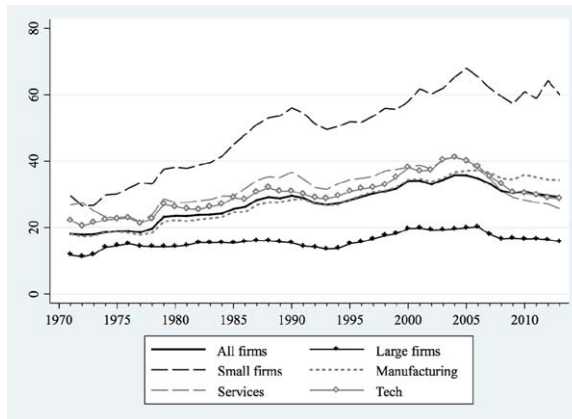

FIGURE 5 Volatility (the coefficient of variation in *S/K*), 1971–2013 [Colour figure can be viewed at wileyonlinelibrary.com]

Note. Volatility is defined as the coefficient of variation (standard deviation relative to the mean) of the firm-level sales to capital ratio, where the standard deviation and mean are based on 5 years of lags. A firm is categorized as small if its total assets are in the bottom quartile of the asset distribution for any given year. A firm is categorized as large if its total assets are in the top quartile of the asset distribution for any given year.
Source. Compustat, author's calculations

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The influence of shareholder value norms on managerial decision-making may, however, be primarily limited to large firms. This distinction is suggested by Figure 3, which indicates that buybacks over the post-1970 period are concentrated among large firms. The fact that repurchases are procyclical implies that firms repurchase shares when stock prices are high, such that small firms are less able to afford large-scale stock repurchases, as compared to large firms. Furthermore, stock option-based pay is greater among large firms, both in absolute value and relative to firm size, than among small firms (Core, Holthausen, & Larcker, 1999).

3 | INVESTMENT

3.1 | Framework

To motivate the econometric specification below, this section outlines firm-level determinants of investment demand, providing a stylized framework with which to analyze the implications of changes in NFC portfolio composition and financing behavior for investment. Consider a firm that invests in two types of assets—fixed capital (K) and financial assets (M)—and that finances its expenditures via a combination of internal funds, new debt (D), and proceeds from new equity issues. Using a dot over a variable to denote a time rate of change, the firm's uses of funds include the acquisition of fixed capital ($\dot{K}=I$) and financial assets ($\dot{M}$), dividend payments to shareholders (Div), and interest payments on outstanding debt ($i^{debt}D$, where i^{debt} is the firm's cost of borrowing). The firm's sources of funds include profits earned on fixed capital (π), returns earned on financial assets ($i^{dep}M$, where i^{dep} is the financial profit rate), new share issues ($\dot{N}$ new shares at a price of ν per share) and new borrowing ($\dot{D}$).

The firm's finance constraint can then be expressed as:

$$pI+\dot{M}+Div+i^{debt}D=\pi+i^{dep}M+\nu\dot{N}+\dot{D} \tag{1}$$

where p denotes the price of the investment good. Equation (1) is an identity, capturing that the firm's total sources of funds must be equivalent to its total uses of funds.[9] This expression highlights the interdependence of investment and financing decisions: the decision to invest in fixed capital or acquire financial assets is concurrent with the decision of how to finance that asset acquisition. Note that repurchases are incorporated by $\dot{N} < 0$.

For given objectives, the firm has desired stocks of capital, financial assets and debt, defined as the levels the firms would select if it could adjust each stock freely and instantaneously in pursuit of its objectives, subject to labor market, demand and financing constraints.[10] In turn, the evolution of the firm's capital stock—that is, the investment decision—can be described by a stock adjustment toward the firm's desired stock of capital. Furthermore, because K, M and D are jointly determined, the evolution of the capital stock depends not only on the discrepancy between the current and desired level of capital, but also on the simultaneous discrepancies between the current and desired levels of financial

[9]Note that Equation (1) is compatible with expansions in a firm's total pool of funds, for example via new borrowing. Thus, Equation (1) does not restrict the possibility that NFCs borrow to repurchase stock.

[10]Of course, firms do not directly maximize objective functions; however, strict maximization is not necessary for the qualitative discussion to hold.

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assets and debt. The adjustment of the firm's capital stock over time ($\dot{K}$) can, therefore, be summarized:[11]

$$\dot{K} = f(K^* - K, M^* - M, D^* - D) \tag{2}$$

It is plausible to assume linear homogeneity, such that a doubling of K^*, for example, leads to a doubling of $\dot{K}$, in which case the firm's accumulation rate can be written:

$$\hat{K} = \frac{\dot{K}}{K} = \frac{I}{K} = f\left(\frac{K^* - K}{K}, \frac{M^* - M}{K}, \frac{D^* - D}{K}\right) \tag{3}$$

$$= f\left(\frac{K^*}{K}, \frac{K^*}{K}\frac{M^*}{K^*}, \frac{M}{K}, \frac{K^*}{K}, \frac{D^*}{K^*}, \frac{D}{K}\right) \tag{4}$$

Investment is, thus, a function of the current and desired stocks of capital, financial assets and debt. While, for given objectives, a complex set of factors influence the determination of the desired stocks (K^*, M^* and D^*), the determination of these stocks is summarized here by the firm's profit rate on fixed capital, financial profit rate and cost of borrowing. A higher profit rate on fixed capital (r) increases the desired stock of capital; however, future profits are unknown, such that investment depends instead on the expected profit rate on new capital (Kalecki, 1971; Robinson, 1956). A higher expected profit rate on fixed capital (r^e) makes holding capital more desirable ($K^*_{r^e} > 0$). Similarly, all else equal, a higher financial profit rate (i^{dep}) increases the desirability of holding financial assets ($M^*_{i^{dep}} > 0$), whereas a higher cost of borrowing (i^{debt}) reduces the desired stock of debt ($D^*_{i^{debt}} < 0$). Due to imperfect competition in goods markets and imperfections in financial markets—requiring, for example, collateral to obtain external financing—the desired stocks are finite. Note, furthermore, that these are firm-specific rates, such that the firm's financing conditions reflect, for example, not only the macroeconomic interest rate environment, but also how the interest rate faced by an individual firm depends on factors such as leverage and wealth.

Thus, assuming that individual NFCs do not have price-setting power in financial markets, the financial profit rate and the cost of borrowing are exogenous, albeit firm-specific, determinants of investment. With imperfect competition in product markets, however, expected profitability is not an exogenous parameter. Expected profitability is, therefore, summarized by the combination of the current profit rate (r) and the utilization rate of fixed capital (u), which together capture the demand and production conditions facing the firm (see Skott, 1989). Because expectations regarding future profits are formed largely on the basis of recent performance, the current profit rate (r) is one indicator of expected future profitability. However, un-utilized capital does not earn profits; thus, the expected return on new capital also depends on whether the additional unit of capital will be utilized (Steindl, 1952). Substituting these variables into Equation (4), Equation (5) summarizes the determinants of investment demand:

[11]The adjustment of the stock of equity is not included separately, because after capital, financial assets and debt, the book value of equity is a residual. The stock adjustment in Equation (2) can, similarly, describe the adjustments of financial assets and debt:

$$\dot{M} = h(K^* - K, M^* - M, D^* - D)$$
$$\dot{D} = z(K^* - K, M^* - M, D^* - D)$$

where these adjustment processes must be jointly satisfied, and the speed of adjustment of the three stocks differs; the speed of adjustment of financial assets may, in particular, approach infinity.

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$$\frac{I}{K} = f\left(r, u, i^{dep}, i^{debt}, M, D, K\right) \tag{5}$$

The expected signs follow from the discussion above. First, the profit rate and the utilization rate are positively related to the investment rate. Second, financial profitability may, on the one hand, capture the 'hurdle' rate of return a manager must expect to earn on fixed capital in order to invest in fixed rather than financial assets, such that the financial profit rate is negatively associated with the investment rate. On the other hand, the relationship may be positive if an increase in financial profitability creates a larger overall pool of funds with which managers can finance the acquisition of both fixed and financial assets. Third, because capital investments are generally financed with a combination of internal and external funds, an increase in the cost of borrowing external funds decreases investment demand at otherwise equal expected rates of return.

Turning to the current stocks of financial assets and debt, first, the stock of financial assets is positively related to the investment rate. This sign reflects that, for given financial conditions, the firm has a desired portfolio composition defined by M^* and K^*. Thus, the relationship between financial assets and capital captures a portfolio adjustment process whereby, for given objectives and at given rates of return on fixed and financial assets, a firm holds financial assets and fixed capital in a relatively stable proportion. If the firm's current financial assets exceed its desired stock, resources are reallocated toward physical capital, and investment rises.

Last, an increase in the stock of debt, all else equal, decreases the investment rate. Debt entails future cash payment commitments, such that a larger stock of debt increases both lenders' and borrowers' risk, reducing the firm's investment demand (Kalecki, 1971; Keynes, 1936; Minsky, 1975). From the perspective of management, a larger stock of debt reduces the firm's margin of safety with which to respond to adverse shocks, thereby decreasing managerial willingness to tie up funds in capital investments. From the perspective of creditors, a large stock of debt signals potential solvency problems and intensifies agency problems in the lending relationship, such that a large stock of debt may inhibit a firm's ability to obtain (additional) external funds, constraining future investment.

3.2 | Shareholder value orientation

For given expected profit rates and cost of borrowing, the desired stocks of capital, financial assets and debt—and, thus, the specific functional form of investment demand—depend on the firm's objectives, which have not been explicitly specified. Importantly, however, growing shareholder value orientation has arguably led to the internalization of 'value maximizing' norms and a shift in objectives toward increasing emphasis on 'value'. It has, similarly, been a frequent claim in the literature that a growing emphasis on shareholder value has shortened managerial time horizons, such that managers targeting 'value' are less likely to tie up funds in long-term, irreversible capital investments than managers targeting 'traditional objectives'.

The implication is that a growing emphasis on 'value maximizing' objectives shifts the investment demand function, with a direct negative effect on the investment rate:

$$\frac{I}{K} = \tilde{f}\left(r, u, i^{dep}, i^{debt}, M, D, K; Sv\right) \tag{6}$$

where Sv denotes shareholder value objectives and $\left(\frac{I}{K}\right)_{Sv} < 0$. Thus, all else equal—that is, at an otherwise equal financial profit rate, expected profit rate, utilization rate, cost of borrowing, and stocks of capital, financial assets and debt—a manager aiming to maximize a firm's stock market valuation

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allocates fewer funds toward long-term capital investment projects than a manager targeting traditional objectives.

Stockhammer (2004) also argues that shareholder value ideology constrains investment via changing managerial preferences. Empirically, Stockhammer equates shareholder value objectives with rentiers' income, a variable similar to financial profits. Because rentiers' income is endogenous to the investment decision, however, it is also interrelated with the firm's other financial decisions, such as the use of debt and equity. Equating shareholder value objectives with rentiers' income, therefore, omits changes in a firm's response to a given cost of borrowing or financial profitability that may accompany an increased emphasis on shareholder value. As such, a relationship between rentiers' income and investment does not clearly identify a link between shareholder value orientation and NFC investment behavior. Here, the implications of shareholder value norms are, instead, explored via a shift in the finance constraint to allow for possible impacts of shareholder value norms on other financial decisions, in addition to investment.

Other factors—including increased international competition and the globalization of production—are also likely to influence NFC investment behavior in this period. Rather than proposing an exhaustive explanation of factors driving changes in investment behavior, however, this article focuses more narrowly on the implications of the particular channels outlined here for domestic investment. The exclusion of other potentially relevant factors is, however, a limitation of this article.

4 | EMPIRICAL STRATEGY AND DATA

4.1 | Statistical specification

The empirical specification of the investment function follows from the discussion in Section 3:

$$\left(\frac{I}{K}\right)_{i,t} = \beta_0 + \beta_1\left(\frac{I}{K}\right)_{i,t-1} + \beta_2 u_{i,t-1} + \beta_3 r_{i,t-1} + \beta_4 i^{dep}_{i,t-1} + \beta_5 i^{debt}_{i,t-1} + \beta_6\left(\frac{M}{A}\right)_{i,t-1} + \beta_7\left(\frac{D}{A}\right)_{i,t-1} + \beta_8 R_{k,t-1} + \beta_9 V_{i,t-1} + \varepsilon_i, t \tag{7}$$

where i denotes firm, t denotes year, k denotes industry and A denotes total assets to capture firm size.[12] The specification utilizes lagged rather than contemporaneous values of the explanatory variables, reflecting that while profits earned in period t are, for example, still unrealized when investment decisions in period t are made, profits in period $t-1$ constitute the most recently observed profits and are, therefore, a determinant of the decision to invest. The investment rate is, similarly, defined as a function of lagged explanatory variables in Orhangazi (2008) and Stockhammer (2004).

Equation (7) includes a lagged dependent variable to capture persistence in capital stock adjustment stemming from the long-term and irreversible nature of capital investments, as well as from adjustment costs in the implementation of new capital. The coefficient on the lagged dependent variable is expected to be positive ($\beta_1 > 0$). The remaining expected signs follow from Section 3. The coefficient on capacity utilization (u) is expected to be positive ($\beta_2 > 0$) and, controlling for capacity utilization, the coefficient on the profit rate is also expected to be positive ($\beta_3 > 0$). The expected sign on the financial profit rate is a priori ambiguous ($\beta_4 \gtrless 0$), and the cost of borrowing is expected to be negatively related

[12]The specification is in levels, rather than log form. While log transformations may account for potential non-linearities in the relationship between investment and the explanatory variables, the incidence of negative explanatory variables implies that log transformations restrict sample size and, more importantly, censor particular types of firms (namely, those earning negative real rates of return). The results are, however, robust to a log-linear specification.

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to fixed investment ($\beta_5 < 0$). The coefficient on the stock of financial assets is expected to be positive ($\beta_6 > 0$) while the coefficient on the outstanding stock of debt is expected to be negative ($\beta_7 < 0$).

The empirical specification also controls for firm-level volatility. For given objectives, a firm's desired stocks of capital, financial assets and debt depend on its environment, and rising volatility over the post-1970 period signals changes in the environment within which NFCs make investment and financing decisions. Like shareholder orientation, the impact of firm-level volatility on the investment decision is, therefore, also conceptualized via a shift in the investment demand function: managers react differently to the same financial variables in a highly volatile versus in a less volatile environment. Specifically, higher volatility reflects greater uncertainty such that, all else equal, a manager facing high volatility is expected to invest less in fixed capital, which is generally long-term and largely irreversible, than a manager facing low volatility. Accordingly, higher firm-level volatility is expected to depress fixed investment ($\beta_9<0$). Volatility is, furthermore, expected to have differential effects on investment for different sized firms, and in particular, to most strongly impact investment rates of small firms.

4.2 | Shareholder value orientation

In the empirical specification, shareholder value orientation is proxied using average yearly industry-level gross stock repurchases relative to total equity ($R_{k,t}$). The variable is constructed to analyze the impact on NFC investment rates of the expectations of (stock) market participants that managers target stock market-based, or 'value based', indicators of firm performance over profit or growth objectives. The expectation is that norms encouraging managerial maximization of market-based value impinge on the allocation of resources for fixed investment ($\beta_8 < 0$). In particular, managers operating in industries in which average repurchases increase, then also face pressure to target financial indicators of performance, because the firms in each industry constitute a comparison group against which managerial performance is evaluated. As a manager in industry k, higher average industry-level repurchases indicate that managers of other firms in your industry are repurchasing stock, thereby, improving financial metrics of performance and also increasing the value of their own stock options. Consequently, as a manager in this industry, you also face pressure to target these financial indicators. Not doing so, first, makes your firm appear undervalued on the stock market relative to your competitors, making it a candidate for corporate takeover and risking your position of authority as a manager. Second, the value of the stock option pay of other managers in your peer group rises relative to your own. The pressure to allocate funds toward financial targets squeezes fixed investment.

Importantly, this variable does not explore the direct effect of an individual firm's decision to repurchase stock on its own investment. The independent inclusion of firm-level repurchases, however, provides little econometrically relevant information about investment, given the bulky and episodic nature of stock repurchase plans. The indicator, furthermore, has the econometric advantage of being exogenous to the individual firm. Note, finally, that repurchases are a clearer indicator of shareholder value orientation than dividend ratios. The shareholder value movement is associated with increased attention to firm performance metrics linked to share *price*, such as return on equity or earnings per share. Furthermore, while total shareholder payouts increase over the post-1970 period, this growth is largely driven by a dramatic, albeit volatile, increase in repurchases (Lazonick, 2009b).

As suggested by the discussion in Section 2, however, shareholder value norms are expected to primarily influence the behavior of large corporations. These large corporations also drive the sector-level trends. Thus, the relationship between the repurchases variable and the investment rate is expected to be negative for large firms and for the full sample; however, the effect is expected to be stronger for large firms.

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4.3 | Estimation strategy

The empirical specification also includes year and firm fixed effects, which capture unobservable time- and firm-specific factors that are relevant for describing a firm's behavior but cannot be explicitly controlled for in the regression—in the case of firm fixed effects, for example, managerial capability. Equation (7) is, then, estimated using the Arellano–Bond Generalized Method of Moments (GMM) difference estimator (Arellano & Bond, 1991). This estimator corrects for bias in the estimate of the lagged dependent variable that is introduced by a correlation between the transformed lagged dependent variable (after first differencing) and the transformed error term (Nickell, 1981). In addition, the estimator is used to correct for bias stemming from the inclusion of regressors subject to reverse causality (namely, those variables appearing in the firm's finance constraint, shown by Equation (1)). These regressors include capacity utilization, the profit rate, the financial profit rate, the cost of borrowing, debt and financial assets. Because Equation (7) specifies investment as a function of lagged values of these variables, each is treated as predetermined, rather than endogenous, with respect to the error term.

Accordingly, the GMM difference estimator instruments the lagged dependent variable and each predetermined variable with lagged levels of the dependent and predetermined variables (beginning from *t-2*), and differences of the strictly exogenous variables. The appropriateness of this difference estimator is corroborated by a simple AR(1) specification, which points to a statistically significant dynamic component in the investment series, but indicates the series does not approach a random walk. In particular, a robust two-step difference GMM estimation, with year dummies and no additional controls, measures an autoregressive coefficient of 0.387 with a standard error of 0.007 (significant at the 1% level). This time series property of the investment data is important, given that if the underlying series were highly persistent, the difference GMM estimator is expected to suffer from a weak instruments problem (wherein lagged levels provide little useful information about future changes in investment).

The two-step Windmeijer (2005) finite sample correction is used to generate asymptotically robust standard errors. To avoid instrument proliferation due to the fact that the instrument set is quadratic in the time dimension, the instrument set for each estimation is restricted such that the number of instruments is less than the number of firms (see Roodman, 2009).[13] The validity of the instrument set is tested using the Sargan test of overidentifying restrictions. Finally, the presence of serial correlation in the errors is tested using a second-order serial autocorrelation test.

The estimations also include additional lags of the explanatory variables. The inclusion of lags is standard in empirical work on investment functions (Fazzari & Mott, 1986; Fazzari, Hubbard, & Peterson, 1988), capturing both gestation lags and expectations lags (Ndikumana, 1999). Because managers act subject to uncertainty and imperfect information, investment decisions are based on expectations regarding the future. These expectations are formed on the basis of recent experience and are, therefore, captured empirically by lags of the explanatory variables. Lags also ameliorate autocorrelation in the errors; the GMM estimator is only valid in the absence of second-order serial correlation in the error term, which would lead to inconsistent estimates. The results presented below include three

[13]The choice of instrument set follows Roodman (2009), which provides the most explicit guidance in the literature regarding the appropriate number of instruments. The Sargan tests, shown with the estimation results below, point to the validity of these instruments. Given the large sample and long-time dimension, it is important to note that this selection rule yields a relatively large instrument set, which may raise concerns about evaluating model specification since the Sargan test is weakened with a large number of instruments. Importantly, however, the rule in Roodman (2009) indicates that what constitutes a large number of instruments is understood relative to sample size. Furthermore, while the time series property of the investment variable does not suggest difference GMM is plagued by a weak instrument problem, it is nonetheless notable that the results are robust to estimation with Roodman (2009) system GMM estimator (Appendix Table A2). Note, finally, that the method of selecting the instrument set described here implies the number of instruments differs with sample size, but also ensures that selection of the instrument set is consistent across subsamples.

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TABLE 2 Descriptive statistics

	All NFCs		1st quartile		2nd quartile		3rd quartile		4th quartile	
	Med	***SD***	**Med**	***SD***	**Med**	***SD***	**Med**	***SD***	**Med**	***SD***
Investment rate	0.20	0.17	0.23	0.21	0.23	0.20	0.21	0.16	0.16	0.12
Capacity utilization	4.91	9.50	7.33	12.25	6.36	10.58	5.27	8.68	2.90	6.32
Profit rate	0.40	1.37	0.31	2.44	0.42	1.61	0.47	0.88	0.34	0.55
Financial profit rate	−0.01	0.04	−0.02	0.05	−0.02	0.04	−0.01	0.04	−0.01	0.04
Cost of borrowing	0.05	0.09	0.06	0.12	0.06	0.11	0.05	0.09	0.05	0.06
Financial assets	0.38	0.21	0.46	0.22	0.43	0.21	0.38	0.20	0.30	0.17
Debt	0.24	0.21	0.19	0.26	0.18	0.21	0.23	0.22	0.29	0.17
Avg. industry repurchases	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Volatility	0.15	0.14	0.24	0.18	0.19	0.16	0.15	0.13	0.11	0.11
Number of observations	55,951		3,350		10,201		16,894		2,481	
Number of firms	6,444		862		2,100		2,784		2,300	
Avg. number of periods	8.7		3.9		4.9		6.1		10.3	

Note. Firm-size quartiles are defined by quartile of the asset distribution, unconditional on available observations for all regression variables. The final three rows denote the number of observations, number of firms and the average number of periods for each sample.
Source. Compustat, author's calculations.

lags of the explanatory variables, although the qualitative conclusions are robust to varying the number of included lags. However, because volatility is constructed using a 5-year moving average and, therefore, incorporates multiple years of information, only the first lag of volatility is included.

4.4 | Data

The sample is an unbalanced annual panel of publicly traded corporations in the United States based on the Compustat database from 1971 to 2013. The sample includes all non-financial firms incorporated in the United States with non-negative sales or total assets, and at least 5 years of observations. To account for outliers, ratios are trimmed at the 1st and 99th percentile; the results are robust to instead winsorizing the sample. Table 2 summarizes descriptive statistics for the full sample and by firm size, where firm size is defined according to quartile of the asset distribution.

The variable definitions are as follows. The investment rate is capital expenditures relative to the beginning-of-period stock of domestic, tangible capital.[14] Capacity utilization is defined as sales relative to the capital stock. The profit rate on fixed capital is defined as profits (gross operating income)

[14]The capital stock is measured at historical cost, but the results are also robust to measuring the capital stock at replacement cost using perpetual inventory calculations. There are limitations to both historical cost and replacement cost measures of the capital stock. While historical cost measures can introduce measurement error, particularly in periods of high inflation, perpetual inventory calculations are sensitive to the choice of parameters like service life (which have been reduced for tax purposes over time; Krippner, 2012). It is, accordingly, notable that the results are robust to the measurement of capital.

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relative to the capital stock. Analogously, the financial profit rate is financial profits relative to the outstanding stock of financial assets. Financial profits are defined as non-operating income; this category includes interest income, dividend income, equity in earnings, foreign exchange adjustments and gains/losses on sales of marketable securities. Importantly, this category excludes extraordinary items, such that it is limited to activities considered part of a firm's normal operations.[15] Financial assets are the sum of cash and short-term investments, current receivables, 'other' investments and advances. The cost of borrowing is the firm's effective interest burden: interest payments relative to total debt. This variable captures factors contributing to a firm's cost of obtaining external finance, such as the firm's credit rating, banking relationships and outstanding lines of credit. Inflation adjustment utilizes the GNP deflator.[16] Shareholder value norms are proxied via the yearly industry average of gross stock repurchases relative to total equity. Finally, volatility is the coefficient of variation in firm-level sales-to-capital ratio, where the mean and *SD* are averaged over the previous 5 years. These definitions are summarized in Table A1 in the Appendix.

5 | RESULTS

Table 3 presents the baseline regression results for the full sample by for size quartile. Long-run multipliers, summarizing the total effect of the three lags of each explanatory variable on investment, are shown in Table 4. Selected robustness analysis is included in the Appendix.[17]

5.1 | Non-financial determinants of investment

Together, the non-financial determinants of investment—the lagged dependent variable, capacity utilization and the profit rate—point to the validity of the regression model. For the full sample of firms, the coefficient on the first lag of the dependent variable is positive and significant, capturing dynamic effects in investment behavior. With the exception of the smallest quartile of firms, this coefficient is also positive for all size sub-samples, and strengthens as firm size increases.[18]

[15]Empirical definitions of financial profits are constrained by data availability. Therefore, it is important to note, first, that other coefficients are insensitive to the inclusion or exclusion of financial profitability (see Appendix Table A3). Second, the results are robust to defining financial profits as interest plus dividend income. This paper utilizes non-operating income because missing observations severely restrict sample size with this alternate definition.

[16]These are pre-tax rates of return. A firm's average tax rate can be calculated, but it is not possible to determine whether those taxes are applied to financial or non-financial income. The extent to which firms are differentially able to avoid taxation, also, discredits attempts to incorporate a firm's tax burden into measured profit rates (tax havens, for example, are likely to be more heavily utilized by large multinational corporations).

[17]In Table A3, explanatory variables are sequentially dropped, showing robustness to variations in the baseline specification. The final column of Table A3 replaces industry-level repurchases with industry-specific time trends. Table A4 shows robustness of the firm size results to two additional definitions of firm size. In this paper, firm size is defined by quartile of the asset distribution, unconditional on available observations for all regressors, such that size reflects a firm's size within the non-financial corporate sector. However, because smaller firms have a higher incidence of missing values, the sample size for larger quartiles exceeds that for smaller quartiles. To show this definition does not produce misleading results, Table A4, first, defines size by quartile of the asset distribution *conditional* on non-missing data for all regressors (such that firm-size subsamples are approximately, although not strictly, equivalent since firms can switch size categories). Table A4, second, defines size by total sales.

[18]The negative coefficient on the lagged dependent variable among the first quartile may suggest greater lumpiness in smaller firms' investments. Specifically, if small firms undertake capital investments less frequently than large firms, there may be less persistence in small firms' investment rates, and possibly even a reversal in expected sign.

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TABLE 3 Estimation results; dependent variable *I/K*

		All NFCs	1st quartile	2nd quartile	3rd quartile	4th quartile
Investment rate	*t-1*	0.239***	−0.147***	0.059***	0.144***	0.332***
		(0.011)	(0.032)	(0.019)	(0.015)	(0.018)
	t-2	−0.015*	−0.152***	−0.096***	−0.054***	−0.025**
		(0.00772)	(0.024)	(0.013)	(0.010)	(0.013)
	t-3	−0.009	−0.088***	−0.026**	−0.040***	−0.015
		(0.006)	(0.022)	(0.011)	(0.009)	(0.010)
Capacity utilization	*t-1*	0.005***	0.0062***	0.007***	0.004***	0.000
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	t-2	0.000	0.002	0.000	0.001	0.000
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	t-3	0.000	0.002**	0.002***	0.001*	0.001*
		(0.005)	(0.001)	(0.001)	(0.000)	(0.001)
Profit rate	*t-1*	0.018***	0.012**	0.009**	0.030***	0.070***
		(0.004)	(0.006)	(0.004)	(0.005)	(0.011)
	t-2	0.0024	0.003	0.0004	0.003	−0.009
		(0.0031)	(0.004)	(0.005)	(0.004)	(0.008)
	t-3	0.002	0.001	−0.003	0.004	−0.012**
		(0.003)	(0.003)	(0.004)	(0.004)	(0.006)
Financial profit rate	*t-1*	−0.004	−0.217	−0.131	0.065	0.150***
		(0.049)	(0.141)	(0.105)	(0.078)	(0.050)
	t-2	−0.010	0.154	0.078	−0.030	−0.029
		(0.049)	(0.145)	(0.104)	(0.075)	(0.047)
	t-3	−0.068**	−0.104	−0.021	−0.029	−0.118***
		(0.030)	(0.123)	(0.064)	(0.050)	(0.031)
Cost of borrowing	*t-1*	0.001	0.048	−0.044	−0.070***	−0.099***
		(0.021)	(0.077)	(0.033)	(0.022)	(0.033)
	t-2	0.005	−0.065	−0.045	0.013	−0.060**
		(0.021)	(0.080)	(0.035)	(0.024)	(0.029)
	t-3	−0.002	0.024	−0.005	−0.030*	0.019
		(0.011)	(0.044)	(0.021)	(0.017)	(0.018)
Financial assets	*t-1*	0.162***	0.356***	0.173***	0.181***	0.125***
		(0.022)	(0.073)	(0.041)	(0.030)	(0.030)
	t-2	0.016	0.190***	0.060	0.062**	0.005
		(0.024)	(0.071)	(0.039)	(0.029)	(0.028)
	t-3	−0.019	0.040	0.000	0.0052	−0.027
		(0.019)	(0.059)	(0.033)	(0.025)	(0.023)
Debt	*t-1*	−0.153***	−0.120*	−0.199***	−0.159***	−0.175***
		(0.021)	(0.067)	(0.039)	(0.026)	(0.028)
	t-2	0.007	−0.056	−0.0316	−0.044*	−0.027
		(0.02)	(0.057)	(0.039)	(0.026)	(0.028)
	t-3	0.026	0.121**	0.001	0.033	0.040*
		(0.018)	(0.054)	(0.034)	(0.021)	(0.022)

(Continues)

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TABLE 3 (Continued)

		All NFCs	1st quartile	2nd quartile	3rd quartile	4th quartile
SV orientation	*t-1*	−0.592***	−0.209	−0.390	−0.321	−0.886***
		(0.155)	(1.244)	(0.433)	(0.250)	(0.183)
	t-2	0.100	−0.380	0.945**	0.269	−0.150
		(0.142)	(1.099)	(0.423)	(0.222)	(0.152)
	t-3	−0.185	0.319	−0.109	0.008	−0.334*
		(0.162)	(1.251)	(0.466)	(0.248)	(0.182)
Volatility	*t-1*	−0.053***	−0.098***	−0.070***	−0.059***	−0.021*
		(0.009)	(0.037)	(0.019)	(0.016)	(0.012)
Observations		55,951	3,350	10,201	16,894	25,506
Firms		6,444	845	2,100	2,784	2,481
Instruments		5,000	734	1,800	2,500	2,300
Firm dummies		Yes	Yes	Yes	Yes	Yes
Year dummies		Yes	Yes	Yes	Yes	Yes
Sargan (*p* value)		0.998	0.977	0.589	0.934	0.388
Second-order auto.		0.743	0.833	0.914	0.149	0.584

Note. Regressions based on the two-step Arellano–Bond difference GMM estimator with Windmeijer (2005) robust standard errors in parentheses. Coefficients for year fixed effects are not reported. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively. For each sub-sample, the instrument set includes lags of the predetermined variables beginning from *t-2*; an identical number of lags for each variable are included in the instrument set, with the instrument set restricted to ensure the number of instruments is less than the number of groups. The test statistics for the Sargan test of over-identifying restrictions and for the Arellano–Bond test of second-order autocorrelation are reported by their *p* values. Variables are defined in Section 4.4 and summarized in Table A1 in the Appendix. Firm size is defined by the unconditional distribution of total assets.
Source. Compustat, author's calculations.

The coefficients on capacity utilization and profitability also have the expected signs in all specifications. Coefficients on both the first lag and the long-run multipliers for capacity utilization are positive and significant for the full sample of firms and each size sub-sample, with the exception of the largest quartile. The magnitude of the short-run relationship between capacity utilization and investment, captured by the coefficient on the first lag, is large: in the full sample, a 1 *SD* increase in capacity utilization implies a 0.28 *SD* increase in the investment rate. The short-run coefficient on the profit rate is also significant with the expected sign, and the long-run coefficients are significant in most samples. The magnitude of the coefficient, however, is smaller: for the full sample, a 1 *SD* in the profit rate implies a 0.15 *SD* increase in the investment rate.

5.2 | Financial determinants of investment

The results highlight the relevance of changes in NFC financial behavior since the early 1970s for fixed investment. Beginning with the financial profit rate, the coefficient on the long-run multiplier for the full sample of firms is negative and weakly significant, although the short-run effect—shown by the first lag of the financial profit rate—is insignificant. The negative long-run coefficient is consistent

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TABLE 4 Long-run coefficients; dependent variable *I/K*

	All NFCs	1st quartile	2nd quartile	3rd quartile	4th quartile
Capacity utilization	0.005***	0.008***	0.009***	0.006***	0.002
	(0.001)	(0.002)	(0.001)	(0.001)	(0.002)
Profit rate	0.028***	0.011*	0.007	0.039***	0.090***
	(0.006)	(0.007)	(0.006)	(0.007)	(0.017)
Financial profit rate	−0.104*	−0.120	−0.070	0.006	0.005
	(0.062)	(0.180)	(0.149)	(0.120)	(0.060)
Cost of borrowing	0.005	0.005	−0.089**	−0.092**	−0.198***
	(0.040)	(0.104)	(0.056)	(0.132)	(0.068)
Financial assets	0.202***	0.423***	0.220***	0.261***	0.145***
	(0.030)	(0.078)	(0.053)	(0.039)	(0.043)
Debt	−0.153***	−0.040	−0.216***	−0.178***	−0.228***
	(0.022)	(0.049)	(0.042)	(0.029)	(0.035)
SV orientation indicator	−0.864**	−0.195	0.421	−0.046	−1.935***
	(0.370)	(1.808)	(0.774)	(0.485)	(0.450)
Volatility	−0.067***	−0.071***	−0.066***	−0.062***	−0.029*
	(0.012)	(0.027)	(0.018)	(0.016)	(0.016)

Note. Long-run coefficients are based on the regression results in Table 3. Long-run coefficients are calculated on the basis of an autoregressive process: the sum of the coefficients on the lags of each variable, divided by one minus the sum of the coefficients on the lags of investment. The *p* values are based on a chi-square statistic. The standard errors, shown in parentheses, are calculated by dividing the estimate by square root of the chi-square statistic. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively.
Source. Compustat, author's calculations.

with Demir (2009), who emphasizes a rising rate of return gap between financial and fixed profitability to be a key determinant of NFCs' portfolio shift away from fixed capital in emerging economies.

Note furthermore that, while the long-run coefficients are insignificant in each size subsample, the short-run effects by firm-size are interesting. In particular, the coefficient on the first lag for the largest quartile of firms is positive and strongly significant, suggesting that an increase in financial profitability supports large firms' investment in the short-run. This result raises the possibility that firms of different sizes hold financial assets for different reasons. As discussed in Section 3, a positive relationship between financial profitability and investment may reflect that financial profits ease financial constraints: in this case, rather than indicating an increase in the 'opportunity cost' of acquiring fixed capital, higher financial profitability increases the firm's total pool of available financing and encourages fixed investment. A related interpretation suggests that large firms may, furthermore, be able to generate complementarities between financial profits and the non-financial components of their business that are not generated by smaller firms. For example, while the financial assets acquired by small firms may hedge against volatility and risk, those of large NFCs may instead reflect movement into the provision of financial services, namely borrowing and lending for profit. This explanation is consistent with anecdotal evidence of large firms' movement into the provision of financial services. NFC expansion into car loans and store-issued credit cards are particularly cogent examples (Froud, Sukhdev,

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Leaver, & Williams, 2005). Store-issued credit cards, for example, generate financial profits and also capture demand for the firm's non-financial products, thereby supporting fixed investment.

The coefficients on the first lag and long-run coefficients of the effective interest burden are negative for firms outside of the first quartile, and strongly statistically significant for firms of above-median firm size. Among the largest quartile of firms, the magnitude of the short run standardized coefficient is 0.05. For the full sample of firms, however, both the first lag and the long-run coefficient are statistically insignificant and, while imprecisely measured, almost zero in standardized terms. Notably, Orhangazi (2008) emphasizes the strength of a negative, and in most subsamples significant, relationship between the first lag of NFC payments to the financial sector and fixed investment. However, Orhangazi's payments variable combines interest payments with shareholders payouts (dividend payments and stock buybacks). The strength of Orhangazi's results relative to these coefficients on the effective interest rate suggests the strength of Orhangazi's finding reflects payouts to shareholders, rather than to creditors.

The stock of financial assets has a positive, statistically significant and strongly robust relationship to fixed investment in both the short-term and the long-run in all specifications. This positive relationship is consistent with the portfolio adjustment process described in Section 3: holding expected returns constant, firms hold both fixed and financial assets, and physical investment increases with the stock of financial assets. Thus, firms acquire financial assets—which ameliorate inherent risks of long-term and irreversible capital investments—concurrently with fixed capital. In the full sample, a 1 *SD* increase in the stock of financial assets is associated with a 0.20 *SD* increase in the investment rate. Furthermore, the magnitude of the coefficient is smaller for large firms, particularly in the short run, which is consistent with the idea that large firms face fewer constraints than small firms in obtaining external finance and, therefore, depend less strongly on the smoothing function of financial assets.

The stock of financial assets has not been included in the empirical literature on financialization and investment, and the independent inclusion of the stock of financial assets in addition to the financial profit rate is an innovation of this article. Importantly, these results do not lend support to the common claim in the financialization literature that NFC acquisition of financial assets has 'crowded out' physical investment, and furthermore suggest that there are certain channels through which financialization may *not* reduce investment, even in the long-term.[19] This result, therefore, points to the importance of delineating specific channels through which a broadly-defined phenomenon of financialization impacts investment: different features of financialization are likely to have differential effects on investment behavior. Note, furthermore, that while the stock of financial assets is found to have a robust positive relationship to fixed investment, the financial profit rate is negatively related to investment in most specifications, suggesting different time implications of the financial profit rate and an acquired stock of financial assets. An increase in the financial profit rate may drive a short-term reallocation of funds toward financial assets, but a larger stock of financial assets provides flexibility to carry out long-term fixed investment projects despite uncertainty regarding future profits or the cost and availability of external finance.

Finally, for the full sample and all size sub-samples, an increase in the stock of debt is found to constrain investment. In both the short run and the long run, debt has a negative and significant relationship to fixed investment (with the exception of an insignificant long-run coefficient for the smallest quartile of firms). In the full sample, a 1 *SD* increase in the stock of debt is associated with a 0.19 *SD* decline in investment. As noted in Section 3, debt finance—by, for example, making the firm more vulnerable to future changes in interest rates, and requiring the firm to set aside a greater proportion of

[19]A specification that instead includes the *change* in financial assets, also, estimates a positive and significant relationship between this *change* in financial assets and investment, corroborating the conclusion that the acquisition of financial assets has *not* directly crowded out fixed capital investment among U.S. firms (see Appendix Table A5).

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future profits—increases inherent risks involved in any investment by increasing both lenders' and borrowers' risk (Kalecki, 1971; Minsky, 1975). Thus, an increase the stock of debt restricts the allocation of funds for capital investment. This finding is consistent with the existing literature, which also highlights a robust negative relationship between a firm's stock of debt and investment rate (see Ndikumana, 1999 for a detailed discussion of the relationship between debt and investment). Importantly, despite a concurrent increase in (liquid) financial assets, gross indebtedness asserts an *independent* effect on balance sheet fragility and, thus, on investment rates. Thus, particularly among large firms, whose stocks of debt rise substantially since 1970, this negative relationship points to a marked decline in the support of external finance for fixed investment in recent decades.

5.3 | Firm-level volatility

The estimations also control for the role of firm-level volatility in explaining post-1970 changes in NFC investment behavior. Firm-level volatility is found to have a negative and significant relationship to fixed investment rates for both the full sample and each subsample of firms; this relationship is particularly strong among the smallest firms, and is only weakly significant among the largest quartile. Across all NFCs, a 1 *SD* increase in volatility is associated with a 0.04 *SD* decline in the investment rate. For the smallest quartile of firms a 1 *SD* increase in volatility is associated with a 0.08 *SD* decline in the investment rate. Among the largest quartile of firms, conversely, a 1 *SD* increase in volatility is associated with only a 0.02 *SD* decline in the investment rate, and the coefficient is only weakly significant. Thus, larger firms are less sensitive to a given increase in volatility than smaller firms, reflecting that large firms—with more total assets and market power—are less vulnerable to swings in sales than small firms. Given that the total increase in volatility is also especially dramatic among small firms, the cumulative effect of rising volatility is particularly important in explaining the investment behavior of small NFCs over the post-1970 period. Because volatility is, similarly, expected to influence the decisions to acquire financial assets and debt—both of which are closely linked to the financialization of NFCs—volatility is an important factor to consider in analyses of the financialization of NFCs.

5.4 | Shareholder value norms

The results, finally, point to a negative relationship between shareholder value norms and fixed investment rates. In the full sample, both the first lag and the long-run multiplier of the repurchases variable are found to have a significant negative relationship with NFC investment rates, implying that an increase in average industry-level repurchases leads to a decline in the investment rates of the firms in that industry. This result suggests that managers in industries in which average repurchases rise also face pressure to target financial performance indicators, and this pressure to reallocate funds toward financial targets squeezes fixed investment. This conclusion is consistent with the literature that emphasizes changes in corporate governance associated with shareholder value ideology as a primary feature of financialization (Lazonick, 2009a; Lazonick & O'Sullivan, 2000). However, these results are the first to draw an explicit link between shareholder value orientation and NFC investment behavior. Orhangazi (2008) finds that total payments from NFCs to the financial sector are negatively correlated with fixed investment rates. As evidence of a negative relationship between shareholder value orientation and investment, however, these findings are only suggestive: the measure conflates payments to creditors and payments to shareholders, and does not explicitly analyze changes in corporate governance associated with shareholder value orientation.

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Tables 3 and 4 also point to firm-size differences: shareholder value norms are found to most significantly impact the investment behavior of the largest quartile of firms. While the estimated coefficient for the first lag is negative for each size sub-sample, the estimates for the first three quartiles of firms are imprecisely measured and statistically insignificant. Accordingly, the statistical significance of the full-sample effect is driven by the largest quartile of firms. The magnitude of the effect among the largest firms is, furthermore, greater than for the full sample, highlighting that the impingement of shareholder value norms on fixed investment is stronger among large corporations. A 1 *SD* increase in average industry-level repurchases is associated with a 0.03 *SD* decline in investment for the full sample of firms, whereas among the largest quartile of firms, a 1 *SD* increase in average industry-level repurchases is associated with a 0.07 *SD* decline in investment.[20] The magnitude of the accumulated long-run effect among large firms is even greater: among the largest quartile of firms, a 1 *SD* increase in average industry-level repurchases is associated with a 0.16 *SD* decline in the investment rate over time.[21]

6 | CONCLUSION

Using a firm-level panel of U.S. corporations, this article presents evidence that shareholder value orientation inhibits the allocation of funds for fixed investment. Simultaneously, certain trends in NFC financial structure—in particular, rising portfolio shares of financial assets—are found to support capital investment. Thus, while some aspects of financialization depress investment, in line with the main contentions in the existing literature on financialization and investment (Demir, 2009; Lazonick, 2009a; Orhangazi, 2008; Stockhammer, 2004; van Treeck, 2008), other trends *support* physical investment rates. These results point to the multi-dimensionality of financialization, and indicate that—because financialization reflects such a large process of structural change—different aspects of financialization may have different implications for firm investment behavior. It is, therefore, important to identify specific channels through which financialization influences investment.

First, shareholder value orientation is found to inhibit the allocation of funds for capital investment between 1971 and 2013. Importantly, growing shareholder orientation is a feature particular to the post-1970 economy that *underlies* the changes in NFC financial structure pointing to the financialization of NFCs. The identification of this trend is important in that it re-orients the discussion from a very broadly-defined phenomenon of 'financialization' to a more specific discussion regarding precise drivers of increasingly financialized behavior among NFCs, and links this underlying driver to investment outcomes.

Shareholder value norms inhibit fixed investment by inducing a shift in managerial priorities toward financial targets, and this effect is particularly strong among large NFCs. This result corroborate

[20]The alternate definitions of firm size used in Appendix Table A4 reiterate these firm-size differences. The long-run coefficients on shareholder orientation are also reported in Table A4.

[21]Appendix Table A6 presents sub-period estimations, which indicate that the relationship between shareholder value norms and fixed investment becomes significant in sub-periods beginning in 1981. This result is consistent with the emphasis in the financialization literature on the increasing relevance of financial motives in the 1980s. In addition, the magnitude of the post-1980 effect substantially exceeds the magnitude of the effect in the full period estimations. For the 1981–2001 period, shown in Column 3 of Table A6, a one standard deviation increase in average industry-level repurchases is associated with a 0.08 standard deviation decline in investment for the full sample—roughly double the magnitude of the full-sample, full-period effect. However, the low p value for the Sargan test in Columns 1–3 of Table A6 indicates that, while these results are suggestive of a notable break after 1980, they should be interpreted with caution.

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claims in a growing literature critical of shareholder value ideology, which has raised concerns regarding the implications of shareholder value norms for a host of key economic variables, including investment. This article, however, links shareholder value orientation to investment in a novel way, by emphasizing the impact of changing corporate governance norms regarding the appropriate allocation of funds for investment behavior, and presents direct firm-level econometric evidence of a negative relationship between shareholder value orientation and investment. The empirical indicator of shareholder value orientation utilized in this article is exogenous to the individual firm and, thus, identifies the effect of shareholder value norms on investment independently from the firms' other financial decisions. Rising firm-level volatility in the post-1970 U.S. economy, similarly, makes managers less willing to tie up funds in long-term, irreversible investment projects, thereby also constraining fixed investment.

The empirical results also highlight, however, that some changes in NFC financial behavior associated with financialization are positively correlated with investment. In particular, this article presents evidence that an increase in the stock of financial assets is positively associated with fixed investment rates. This result calls into question a narrative in which financialization and, in particular the acquisition of financial assets, 'crowds out' fixed capital. It is important to note, however, that increased holdings of financial assets, while supporting investment rates on average, may nonetheless contribute to increased *volatility* of investment; distinguishing the relationship between financialization and investment rates from the relationship between financialization and investment volatility is an important topic for future research. Finally, the results in this article suggest a positive short-run relationship between financial profitability and investment for large firms, indicating that large firms may exploit complementarities between financial and non-financial activities that unavailable to smaller firms.

Both the descriptive and econometric analysis in this article emphasize firm-size differences, which point to two different stories of financialization for large and for small firms, and suggest that the constraints facing small and large firms have evolved differently over the post-1970 period. As compared to larger firms, the dramatic increase in volatility facing small firms suggests that increasing uncertainty is particularly important for explaining small NFCs' post-1970 changes in financial structure. Concurrent de-leveraging and a declining share of capital in small firm's portfolios, similarly, suggest that small firms have faced growing real-side constraints that have led them to borrow less, hold more liquidity, and invest less in fixed capital. Conversely, while volatility has also impacted large firms, shareholder value orientation has more significantly impacted the investment behavior of large firms, reflecting that large non-financial companies are increasingly oriented toward financial measures of firm performance.

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APPENDIX

TABLE A1 Summary of variable definitions

Variable	Definition	Compustat items
Investment rate	Capital expenditures relative to the capital stock (net property, plant and equipment).	Capital expenditures (128), Capital stock (141)[a]
Capacity utilization	Sales relative to the capital stock.	Sales (12), Capital stock (141)
Profit rate	Gross operating income relative to the capital stock.	Operating income before depreciation (13), Capital stock (141)
Financial profit rate	Gross non-operating income relative to financial assets.[b]	Non-operating income (61), Financial assets (see below)
Effective interest burden	Interest payments relative to total debt (the sum of current and long-term debt).	Interest payments (15), Total debt (34 and 142)
Financial assets	Cash and short-term investments, current receivables, other current assets (less inventories) and 'other' investments and advances (which includes, for example, investments in and advances to unconsolidated subsidiaries and affiliates, and banks and savings & loan investment securities, and miscellaneous assets such as stock or debt issuance costs). (Relative to total assets in the econometric specification).	Financial assets (1, 2, 68, 31, 32 and 69, respectively)
Total debt	Current and long-term debt. (Relative to total assets in the econometric specification).	Total debt (34 and 142)
Industry average of gross repurchases	Gross repurchases relative to total equity.	Gross repurchases (115), Total equity (144)
Coefficient of variation in sales to capital	The standard deviation of S/K relative to the mean; 5-year average.[c]	Sales (12), Capital stock (141)

Inflation adjustment based off the GNP deflator.

[a]This measure of the capital stock reflects the book value of capital (at historical cost). The results are robust to defining the capital stock based on perpetual inventory calculations, such that the capital stock for firm i in period t is $K_{it}=(1-\delta_{it})K_{it-1}+I_{it}$, where I_{it} is new investment, $\delta_{it}=\text{depreciation}_{it}/K_{it}$ (depreciation is item 14) and the first observation of the book value of capital is used as the initial replacement cost estimate for the capital stock for each firm i.

[b]The results are robust to financial profits defined as the sum of interest and dividend income.

[c]The results are robust to defining the coefficient of variation in profits to capital ratio.

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TABLE A2 System GMM estimator; dependent variable *I/K*

		Full sample	1st quartile	2nd quartile	3rd quartile	4th quartile
Investment rate	*t-1*	0.329***	0.206***	0.285***	0.323***	0.496***
		(0.00953)	(0.0314)	(0.0169)	(0.0143)	(0.0148)
	t-2	0.00426	0.0199	−0.0169	0.00157	0.0237**
		(0.00745)	(0.0274)	(0.0145)	(0.0112)	(0.0120)
	t-3	0.00927	0.0321	0.00481	−3.48e-06	0.0309***
		(0.00663)	(0.0328)	(0.0120)	(0.00955)	(0.00957)
Capacity utilization	*t-1*	0.00379***	0.00498***	0.00408***	0.00102**	0.000568
		(0.000350)	(0.000754)	(0.000566)	(0.000513)	(0.000637)
	t-2	−0.00212***	−0.00248***	−0.00191***	−0.000347	7.42e-05
		(0.000393)	(0.000908)	(0.000653)	(0.000532)	(0.000720)
	t-3	0.000252	0.000101	0.000424	0.000294	0.000278
		(0.000309)	(0.000962)	(0.000499)	(0.000483)	(0.000536)
Profit rate	*t-1*	0.0194***	0.0100***	0.0122***	0.0415***	0.0612***
		(0.00242)	(0.00385)	(0.00319)	(0.00545)	(0.00621)
	t-2	−0.00277	0.000840	−0.00994***	−0.00869*	−0.0228***
		(0.00204)	(0.00311)	(0.00343)	(0.00489)	(0.00548)
	t-3	−0.000737	−0.00263	9.56e-05	−0.00320	−0.0133***
		(0.00195)	(0.00361)	(0.00323)	(0.00498)	(0.00432)
Financial profit rate	*t-1*	0.0184	−0.00545	−0.0983*	0.0321	0.0400*
		(0.0212)	(0.116)	(0.0566)	(0.0429)	(0.0212)
	t-2	−0.0258	0.0792	−0.0234	0.000459	−0.0546***
		(0.0187)	(0.0853)	(0.0515)	(0.0373)	(0.0175)
	t-3	−0.0771***	0.0193	−0.0982*	−0.0240	−0.0647***
		(0.0178)	(0.149)	(0.0523)	(0.0358)	(0.0159)
Cost of borrowing	*t-1*	−0.0166	0.0756**	−0.0225	−0.0324*	−0.0641***
		(0.0113)	(0.0360)	(0.0203)	(0.0173)	(0.0160)
	t-2	0.0330***	0.0252	0.000960	0.0624***	0.0160
		(0.0115)	(0.0394)	(0.0219)	(0.0173)	(0.0144)
	t-3	0.0321***	0.0841	0.0300	0.0227	0.0359***
		(0.0101)	(0.0524)	(0.0189)	(0.0149)	(0.0139)
Change in financial assets	*t-1*	0.144***	0.295***	0.154***	0.140***	0.0759***
		(0.0112)	(0.0447)	(0.0236)	(0.0173)	(0.0121)
	t-2	0.0101	−0.0298	0.0288	0.0103	−0.00810
		(0.0102)	(0.0401)	(0.0232)	(0.0176)	(0.0124)
	t-3	−0.0159*	−0.00979	−0.0247	−0.0170	−0.0173*
		(0.00900)	(0.0578)	(0.0201)	(0.0152)	(0.0101)
Debt	*t-1*	−0.138***	−0.0913***	−0.182***	−0.152***	−0.119***
		(0.0104)	(0.0295)	(0.0226)	(0.0163)	(0.0122)
	t-2	0.0331***	0.0364	0.0471**	0.0470***	0.0155
		(0.00950)	(0.0398)	(0.0217)	(0.0159)	(0.0114)
	t-3	0.0445***	0.0958**	0.0853***	0.0178	0.0420***

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TABLE A2 (Continued)

		Full sample	1st quartile	2nd quartile	3rd quartile	4th quartile
		(0.00848)	(0.0411)	(0.0185)	(0.0139)	(0.0100)
SV orientation	*t-1*	−0.428***	−0.309	−0.250	−0.239	−0.613***
		(0.137)	(1.091)	(0.398)	(0.215)	(0.143)
	t-2	0.0955	−0.770	0.711	−0.0471	0.0828
		(0.148)	(1.243)	(0.463)	(0.249)	(0.152)
	t-3	−0.0547	1.407	−0.700*	−0.376*	0.0646
		(0.145)	(1.199)	(0.417)	(0.224)	(0.137)
Volatility	*t-1*	−0.00119	−0.0403**	0.00114	−0.0215	0.00978
		(0.00602)	(0.0206)	(0.0115)	(0.0159)	(0.00618)
Observations		63,884	4,297	12,226	19,486	27,875
Firms		7,359	1,122	2,485	3,186	2,681
Instruments		5,339	831	1,920	2,739	2,634
Firm dummies		Yes	Yes	Yes	Yes	Yes
Year dummies		Yes	Yes	Yes	Yes	Yes
Sargan (*p* value)		0.298	0.940	0.613	0.237	0.337
Second-order auto.		0.232	0.758	0.723	0.055	0.763

Note. Regressions based on the two-step Arellano–Bond system GMM estimator with Windmeijer (2005) robust standard errors in parentheses. Coefficients for year fixed effects are not reported. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively. For each sub-sample, the instrument set includes lags of the predetermined variables beginning from *t-2*; an identical number of lags for each variable are included in the instrument set, with the instrument set restricted to ensure the number of instruments is less than the number of groups. The test statistics for the Sargan test of over-identifying restrictions and for the Arellano–Bond test of second-order autocorrelation are reported by their *p* values. Variables are defined in Section 4.4 and summarized in Table A1 in the Appendix. Firm size is defined by the unconditional distribution of total assets.
Source. Compustat, author's calculations.

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TABLE A3 Sensitivity analysis; dependent variable *I/K*

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Investment rate	*t-1*	0.253***	0.263***	0.244***	0.256***	0.260***	0.276***	0.249***	0.254***	0.267***	0.246***	0.239***	0.240***
		(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.010)	(0.011)	(0.011)
	t-2	−0.013*	−0.005	−0.013	−0.010	−0.006	0.004	−0.017**	−0.016**	−0.020**	0.000	−0.014*	−0.014*
		(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)	(0.008)	(0.008)
	t-3	−0.009	−0.004	−0.009	0.000	0.004	0.004	−0.009	−0.007	−0.009	0.005	−0.008	−0.009
		(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.007)	(0.005)	(0.006)	(0.006)
Capacity Utilization	*t-1*		0.005***	0.005***	0.005***	0.009***	0.005***	0.005***	0.005***	0.005***	0.004***	0.005***	0.005***
			(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	t-2		0.000	0.000	−0.001	−0.002***	0.000	0.000	−0.001	0.000	0.000	0.000	0.000
			(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	t-3		−0.004	−0.001	0.002	0.001	0.000	0.001	−0.001	−0.001	0.000	0.000	0.000
			(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.001)
Profit rate	*t-1*	0.028***		0.017***	0.014***	0.014***		0.018***	0.019***	0.020***	0.015***	0.018***	0.017***
		(0.005)		(0.004)	(0.003)	(0.004)		(0.005)	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)
	t-2	0.001		0.002	0.0001	−0.001		0.001	0.004	0.004	−0.002	0.002	0.0023
		(0.003)		(0.003)	(0.003)	(0.003)		(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
	t-3	0.001		0.002	0.001	0.001		0.002	0.003	0.003	0.000	0.002	0.002
		(0.003)		(0.003)	(0.002)	(0.002)		(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)
Financial profit rate	*t-1*	0.038	−0.046		0.032			0.009	0.012	0.006	0.021	−0.003	0.018
		(0.052)	(0.052)		(0.059)			(0.055)	(0.055)	(0.061)	(0.053)	(0.051)	(0.053)
	t-2	−0.026	0.013		−0.031			−0.025	−0.020	−0.038	−0.022	0.002	0.018
		(0.052)	(0.052)		(0.055)			(0.054)	(0.054)	(0.060)	(0.051)	(0.050)	(0.050)
	t-3	−0.055*	−0.065**		−0.063*			−0.083***	−0.066**	−0.074**	−0.050*	−0.075**	−0.076**
		(0.032)	(0.031)		(0.032)			(0.031)	(0.032)	(0.034)	(0.029)	(0.031)	(0.031)
Cost of borrowing	*t-1*	0.022	−0.008	−0.002				0.009	0.051**	0.069***	−0.004	0.001	0.004
		(0.025)	(0.024)	(0.023)				(0.024)	(0.021)	(0.023)	(0.021)	(0.022)	(0.022)
	t-2	0.009	0.012	0.004				0.007	0.026	0.032	0.010	0.003	0.001

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TABLE A3 (Continued)

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
		(0.024)	(0.022)	(0.023)				(0.023)	(0.021)	(0.024)	(0.022)	(0.022)	(0.022)
	t-3	−0.009	0.002	−0.002				−0.002	0.002	0.002	0.006	−0.002	−0.001
		(0.012)	(0.012)	(0.011)				(0.012)	(0.011)	(0.012)	(0.010)	(0.011)	(0.011)
Financial assets	*t-1*	0.193***	0.170***	0.170***	0.195***	0.230***	0.213***		0.188***		0.188***	0.163***	0.154***
		(0.026)	(0.024)	(0.024)	(0.026)	(0.031)	(0.028)		(0.024)		(0.023)	(0.023)	(0.023)
	t-2	−0.004	0.012	0.018	−0.017	−0.026	−0.017		0.019		0.003	0.017	0.011
		(0.027)	(0.025)	(0.026)	(0.026)	(0.030)	(0.030)		(0.026)		(0.024)	(0.025)	(0.025)
	t-3	−0.025	−0.023	−0.019	−0.012	−0.006	−0.025		−0.022		−0.031*	−0.022	−0.030
		(0.021)	(0.019)	(0.019)	(0.020)	(0.022)	(0.023)		(0.020)		(0.018)	(0.019)	(0.019)
Debt	*t-1*	−0.164***	−0.191***	−0.156***	−0.182***	−0.199***	−0.208***	−0.175***			−0.175***	−0.158***	−0.153***
		(0.027)	(0.025)	(0.023)	(0.022)	(0.026)	(0.027)	(0.022)			(0.020)	(0.022)	(0.022)
	t-2	0.013	0.025	0.012	−0.005	0.004	0.0046	0.007			0.014	0.008	0.006
		(0.026)	(0.024)	(0.026)	(0.025)	(0.028)	(0.029)	(0.025)			(0.022)	(0.024)	(0.023)
	t-3	0.026	0.038**	0.025	0.0399**	0.044**	0.050**	0.033*			0.037**	0.026	0.028
		(0.021)	(0.019)	(0.020)	(0.019)	(0.022)	(0.022)	(0.019)			(0.015)	(0.018)	(0.018)
SV orientation	*t-1*	−0.587***	−0.648***	−0.626***	−0.658***	−0.682***	−0.692***	−0.601***	−0.569***	−0.550***	−0.566***		
		(0.164)	(0.162)	(0.161)	(0.158)	(0.169)	(0.164)	(0.156)	(0.159)	(0.159)	(0.154)		
	t-2	0.0265	0.067	0.095	0.081	0.145	0.035	0.127	0.086	0.104	0.171		
		(0.150)	(0.148)	(0.146)	(0.141)	(0.146)	(0.145)	(0.147)	(0.147)	(0.151)	(0.143)		
	t-3	−0.179	−0.191	−0.223	−0.092	−0.134	−0.086	−0.194	−0.153	−0.158	−0.081		
		(0.173)	(0.167)	(0.165)	(0.162)	(0.173)	(0.169)	(0.163)	(0.166)	(0.167)	(0.157)		
Volatility	*t-1*	−0.030***	−0.049***	−0.055***	−0.050***	−0.076***	−0.048***	−0.046***	−0.059***	−0.052***		−0.053***	−0.055***
		(0.009)	(0.010)	(0.010)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)		(0.010)	(0.010)
Observations		55,951	55,993	55,973	64,894	64,916	64,980	55,951	55,951	55,951	55,951	72,553	72,553
Firms		6,444	6,446	6,444	7,248	7,248	7,251	6,444	6,444	6,444	6,444	9,071	9,071

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TABLE A3 (Continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Instruments	4,400	4,300	4,300	4,400	3,900	3,200	4,300	4,300	3,800	4,800	4,800	4,900
Firm dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind. time trends	No	No	No	No	No	No	No	No	No	No	No	Yes
Sargan p value	0.944	0.734	0.586	0.574	0.625	0.146	0.588	0.412	0.638	0.928	0.860	0.915
Second-order auto.	0.645	0.890	0.764	0.476	0.525	0.779	0.752	0.674	0.645	0.752	0.896	0.515

Note. Regressions based on the two-step Arellano–Bond difference GMM estimator with Windmeijer (2005) robust standard errors in parentheses. Coefficients for year fixed effects are not reported. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively. The instrument set includes lags of the predetermined variables beginning from t-2, with an identical number of lags for each variable are included in the instrument set. The test statistics for the Sargan test of over-identifying restrictions and for the Arellano–Bond test of second-order autocorrelation are reported by their p values. Variables are defined in Section 4.4 and summarized in Table A1 in the Appendix. Firm size is defined according to total assets. Column 1 drops capacity utilization; 2–6 drop combinations of the rates of return; 7–9 drop the stocks of financial assets and debt; 10–11 drop average industry repurchases and volatility; 12 replaces repurchases with industry-level time trends.
Source. Compustat, author's calculations.

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TABLE A4 Alternative definitions of firm size; dependent variable *I/K*

		Panel A: Size defined by distribution of total assets, conditional on non-missing observations				Panel B: Size defined by the distribution of revenue earned from sales			
		1st quartile	2nd quartile	3rd quartile	4th quartile	1st quartile	2nd quartile	3rd quartile	4th quartile
Investment rate	*t-1*	0.024	0.115***	0.163***	0.394***	−0.020	0.076***	0.167***	0.330***
		(0.019)	(0.020)	(0.022)	(0.023)	(0.024)	(0.018)	(0.016)	(0.016)
	t-2	−0.095***	−0.054***	−0.072***	−0.037**	−0.117***	−0.065***	−0.062***	−0.026**
		(0.014)	(0.014)	(0.014)	(0.015)	(0.018)	(0.013)	(0.011)	(0.011)
	t-3	−0.034***	−0.042***	−0.042***	−0.012	−0.047***	−0.017	−0.068***	−0.009
		(0.011)	(0.012)	(0.011)	(0.012)	(0.014)	(0.010)	(0.010)	(0.008)
Capacity utilization	*t-1*	0.007***	0.005***	0.002*	−0.002	0.0073***	0.007***	0.004***	0.000
		(0.001)	(0.001)	(0.001)	(0.001)	(0.0013)	(0.001)	(0.001)	(0.001)
	t-2	0.001	0.000	0.001	0.001	0.002*	−0.001	−0.001	0.001**
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	t-3	0.002**	0.001	0.001	0.002	0.002	0.003***	0.001	0.001**
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.0005)	(0.001)
Profit rate	*t-1*	0.006	0.037***	0.042***	0.065***	0.005	0.024***	0.028***	0.058***
		(0.004)	(0.008)	(0.010)	(0.015)	(0.005)	(0.005)	(0.0052)	(0.007)
	t-2	0.003	−0.003	0.012	−0.017	0.002	−0.003	0.009*	−0.009
		(0.003)	(0.008)	(0.011)	(0.011)	(0.004)	(0.005)	(0.005)	(0.007)
	t-3	0.002	−0.001	0.004	−0.014**	0.001	−0.002	0.003	−0.007
		(0.003)	(0.006)	(0.009)	(0.0061	(0.003)	(0.004)	(0.005)	(0.005)
Financial profit rate	*t-1*	−0.084	0.160	0.092	0.134**	−0.038	−0.103	0.031	0.183***
		(0.102)	(0.106)	(0.073)	(0.059)	(0.136)	(0.091)	(0.065)	(0.048)
	t-2	0.070	−0.007	−0.048	−0.158***	0.048	−0.061	−0.071	−0.114**
		(0.102)	(0.100)	(0.070)	(0.053)	(0.128)	(0.094)	(0.065)	(0.045)
	t-3	−0.037	−0.055	−0.143***	−0.015	−0.126	0.020	−0.112**	−0.049*

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TABLE A4 (Continued)

		Panel A: Size defined by distribution of total assets, conditional on non-missing observations				Panel B: Size defined by the distribution of revenue earned from sales			
		1st quartile	2nd quartile	3rd quartile	4th quartile	1st quartile	2nd quartile	3rd quartile	4th quartile
		(0.067)	(0.059)	(0.052)	(0.036)	(0.084)	(0.053)	(0.045)	(0.005)
Cost of borrowing	*t-1*	0.035	−0.055*	−0.076**	−0.061*	0.019	−0.006	−0.091***	−0.073***
		(0.037)	(0.031)	(0.033)	(0.033)	(0.055)	(0.030)	(0.026)	(0.025)
	t-2	−0.014	0.044	0.0071	−0.048	−0.022	0.021	0.014	−0.022
		(0.043)	(0.035)	(0.027)	(0.034)	(0.061)	(0.032)	(0.028)	(0.023)
	t-3	0.012	−0.050**	−0.005	0.031	−0.002	−0.011	−0.037**	0.038**
		(0.023)	(0.020)	(0.020)	(0.020)	(0.033)	(0.020)	(0.017)	(0.020)
Financial assets	*t-1*	0.265***	0.145***	0.160***	0.078**	0.264***	0.176***	0.214***	0.081***
		(0.044)	(0.044)	(0.039)	(0.033)	(0.056)	(0.038)	(0.032)	(0.025)
	t-2	0.084*	0.062	−0.003	0.016	0.153***	0.094***	0.029	0.027
		(0.045)	(0.040)	(0.032)	(0.036)	(0.055)	(0.036)	(0.029)	(0.027)
	t-3	0.050	0.004	−0.012	−0.007	0.049	−0.044	0.017	−0.040
		(0.036)	(0.036)	(0.029)	(0.026)	(0.045)	(0.032)	(0.024)	(0.020)
Debt	*t-1*	−0.135***	−0.184***	−0.129***	−0.155***	−0.082	−0.175***	−0.141***	−0.127***
		(0.043)	(0.036)	(0.032)	(0.031)	(0.052)	(0.033)	(0.028)	(0.022)
	t-2	−0.037	−0.004	−0.048	−0.031	−0.078	−0.057	0.012	−0.045**
		(0.042)	(0.038)	(0.031)	(0.036)	(0.050)	(0.038)	(0.025)	(0.033)
	t-3	0.077**	0.006	−0.003	0.027	0.086*	0.0073	−0.031	0.050***
		(0.037)	(0.030)	(0.028)	(0.025)	(0.047)	(0.030)	(0.023)	(0.024)
SV orientation	*t-1*	−0.119	−0.733*	−0.602**	−0.888***	−0.315	−0.828**	−0.800***	−0.511***
		(0.529)	(0.387)	(0.260)	(0.218)	(0.655)	(0.421)	(0.251)	(0.143)
	t-2	0.668	0.684**	−0.164	−0.207	0.095	0.908**	0.090	−0.186
		(0.493)	(0.332)	(0.227)	(0.184)	(0.630)	(0.361)	(0.233)	(0.142)
	t-3	0.182	−0.166	0.108	−0.693***	0.024	−0.499	−0.120	−0.207

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TABLE A4 (Continued)

	Panel A: Size defined by distribution of total assets, conditional on non-missing observations				Panel B: Size defined by the distribution of revenue earned from sales			
	1st quartile	2nd quartile	3rd quartile	4th quartile	1st quartile	2nd quartile	3rd quartile	4th quartile
	(0.553)	(0.383)	(0.270)	(0.216)	(0.746)	(0.425)	(0.250)	(0.158)
Volatility *t-1*	−0.071***	−0.077***	−0.041**	−0.001	−0.093***	−0.051***	−0.063***	−0.006
	(0.019)	(0.020)	(0.017)	(0.0133)	(0.025)	(0.018)	(0.017)	(0.010)
SV orientation	0.662	−0.220	−0.692	−2.733***	−0.166	−0.416	−0.862*	−1.238***
(Long-run coefficient)	(0.926)	(0.714)	(0.481)	(0.606)	(1.136)	(0.807)	(0.495)	(0.452)
Observations	11,043	12,907	15,032	16,969	6,819	11,652	16,005	21,475
Firms	2,296	2,523	2,224	1,581	1,533	2,350	2,541	2,082
Instruments	1,500	1,500	1,500	1,500	1,100	2,000	2,300	2,000
Firm dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sargan (*p* value)	0.263	0.115	0.351	0.609	0.384	0.623	0.733	0.655
Second-order auto.	0.597	0.613	0.808	0.917	0.866	0.942	0.493	0.945

Note. Regressions based on the two-step Arellano–Bond difference GMM estimator with Windmeijer (2005) robust standard errors in parentheses. Coefficients for year fixed effects are not reported. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively. The objective of the firm-size definition in Panel A is to equalize sample size; therefore, an identical instrument set is used for each sub-sample, with lags of the predetermined variables from *t-2* to *t-7*. In Panel B, the instrument set for each sub-sample includes lags of the predetermined variables beginning from *t-2*; an identical number of lags for each variable are included in the instrument set, with the instrument set restricted to ensure the number of instruments is less than the number of groups. The test statistics for the Sargan test of over-identifying restrictions and for the Arellano–Bond test of second-order autocorrelation are reported by their *p* values. Variables are defined in Section 4.4 and summarized in Table A1 in the Appendix. Firm size is defined in Panel A by total assets, conditional on non-missing values for the variables in the regression specification, and in Panel B by revenue from sales. Long-run coefficients for shareholder value orientation are also reported; see details in caption of Table 4.
Source. Compustat, author's calculations.

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TABLE A5 The change in financial assets; dependent variable *I/K*

		All NFCs	1st quartile	2nd quartile	3rd quartile	4th quartile
Investment rate	*t-1*	0.232***	−0.126***	0.061***	0.139***	0.317***
		(0.083)	(0.041)	(0.020)	(0.016)	(0.016)
	t-2	−0.018	−0.116***	−0.104***	−0.059***	−0.036***
		(0.049)	(0.026)	(0.013)	(0.011)	(0.011)
	t-3	−0.013	−0.065***	−0.029***	−0.051***	−0.028***
		(0.033)	(0.024)	(0.011)	(0.009)	(0.009)
Capacity utilization	*t-1*	0.005***	0.008***	0.007***	0.005***	0.003
		(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
	t-2	−0.000	0.003	0.000	0.001	0.001
		(0.001)	(0.002)	(0.001)	(0.001)	(0.001)
	t-3	−0.000	0.002	0.002***	0.001**	0.002***
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Profit rate	*t-1*	0.012**	0.008	0.005	0.023***	0.061***
		(0.005)	(0.008)	(0.006)	(0.005)	(0.006)
	t-2	0.002	0.005	0.001	0.000	−0.014**
		(0.003)	(0.006)	(0.005)	(0.005)	(0.006)
	t-3	0.001	0.001	−0.001	0.003	−0.012**
		(0.003)	(0.005)	(0.005)	(0.004)	(0.005)
Financial profit rate	*t-1*	−0.002	−0.133	−0.152	0.207***	0.202***
		(0.060)	(0.267)	(0.098)	(0.076)	(0.047)
	t-2	−0.070	0.065	0.081	−0.017	−0.083*
		(0.086)	(0.193)	(0.101)	(0.073)	(0.043)
	t-3	−0.047	−0.086	−0.028	−0.056	−0.131***
		(0.031)	(0.151)	(0.063)	(0.049)	(0.028)
Cost of borrowing	*t-1*	0.005	0.076	−0.056*	−0.054***	−0.073***
		(0.048)	(0.086)	(0.033)	(0.021)	(0.026)
	t-2	0.004	−0.065	−0.045	0.025	−0.049**
		(0.068)	(0.089)	(0.034)	(0.023)	(0.022)
	t-3	−0.010	0.005	−0.001	−0.032*	0.012
		(0.071)	(0.054)	(0.021)	(0.016)	(0.014)
Change in financial assets	*t-1*	0.159***	0.230***	0.136***	0.148***	0.166***
		(0.017)	(0.058)	(0.024)	(0.019)	(0.017)
	t-2	0.046***	0.141**	0.050*	0.072***	0.052***
		(0.017)	(0.060)	(0.027)	(0.020)	(0.016)
	t-3	0.007	0.057**	0.005	0.015	0.005
		(0.008)	(0.027)	(0.013)	(0.011)	(0.007)
Debt	*t-1*	−0.164***	−0.094	−0.190***	−0.170***	−0.174***
		(0.032)	(0.073)	(0.037)	(0.025)	(0.021)
	t-2	0.005	−0.147*	−0.044	−0.008	−0.013
		(0.058)	(0.080)	(0.036)	(0.026)	(0.022)
	t-3	0.039	0.230***	0.006	0.013	0.038**
		(0.103)	(0.065)	(0.034)	(0.022)	(0.019)

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TABLE A5 (Continued)

		All NFCs	1st quartile	2nd quartile	3rd quartile	4th quartile
SV orientation	*t-1*	−0.522**	0.536	−0.590	−0.163	−0.622***
		(0.211)	(1.363)	(0.446)	(0.258)	(0.143)
	t-2	0.122	1.278	1.004**	0.217	−0.129
		(0.165)	(1.165)	(0.429)	(0.233)	(0.130)
	t-3	−0.250	0.519	−0.567	−0.188	−0.231
		(0.201)	(1.437)	(0.454)	(0.254)	(0.153)
Volatility	*t-1*	−0.064***	−0.108**	−0.084***	−0.071***	−0.025**
		(0.012)	(0.043)	(0.019)	(0.016)	(0.010)
Observations		52,988	3,059	9,512	16,018	24,399
Firms		6,262	790	1,980	2,708	2,438
Instruments		4,800	612	1,800	2,500	2,400
Firm dummies		Yes	Yes	Yes	Yes	Yes
Year dummies		Yes	Yes	Yes	Yes	Yes
Sargan (*p* value)		0.939	0.495	0.868	0.949	0.821
Second-order auto.		0.824	0.415	0.881	0.291	0.366

Note. Regressions based on the two-step Arellano–Bond difference GMM estimator with Windmeijer (2005) robust standard errors in parentheses. Coefficients for year fixed effects are not reported. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively. For each sub-sample, the instrument set includes lags of the predetermined variables beginning from *t-2*; an identical number of lags for each variable are included in the instrument set, with instrument set is restricted to ensure the number of instruments is less than the number of groups. The test statistics for the Sargan test of over-identifying restrictions and for the Arellano–Bond test of second-order autocorrelation are reported by their *p* values. Variables are defined in Section 4.4 and summarized in Table A1 in the Appendix. The change in financial assets is measured as the first difference in firm-level financial assets, relative to total assets. Firm size is defined according to quartile of the sales distribution.
Source. Compustat, author's calculations.

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TABLE A6 Rolling sub-period estimations; dependent variable *I/K*

		1971–1990	1976–1995	1981–2000	1986–2005	1991–2010
Investment rate	*t-1*	0.238***	0.227***	0.240***	0.234***	0.221***
		(0.019)	(0.016)	(0.015)	(0.014)	(0.014)
	t-2	0.012	0.007	0.003	−0.021**	−0.026***
		(0.014)	(0.012)	(0.011)	(0.010)	(0.010)
	t-3	0.005	0.000	−0.001	−0.007	−0.019**
		(0.009)	(0.008)	(0.008)	(0.008)	(0.009)
Capacity utilization	*t-1*	0.005***	0.006***	0.005***	0.005***	0.004***
		(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
	t-2	0.000	0.0003	0.000	−0.001*	0.000
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
	t-3	0.000	−0.001	−0.001	0.000	−0.001
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Profit rate	*t-1*	0.012	0.013	0.015**	0.020***	0.023***
		(0.011)	(0.009)	(0.007)	(0.007)	(0.004)
	t-2	−0.001	0.001	0.000	0.002	0.006
		(0.005)	(0.005)	(0.004)	(0.004)	(0.004)
	t-3	−0.001	−0.005	0.001	0.004	0.002
		(0.005)	(0.005)	(0.004)	(0.004)	(0.004)
Financial profit rate	*t-1*	0.006	−0.017	−0.084	−0.012	−0.069
		(0.091)	(0.067)	(0.068)	(0.059)	(0.060)
	t-2	−0.120	−0.069	−0.081	−0.062	−0.040
		(0.091)	(0.075)	(0.069)	(0.059)	(0.055)
	t-3	−0.017	−0.021	−0.019	−0.045	−0.025
		(0.050)	(0.042)	(0.041)	(0.035)	(0.036)
Cost of borrowing	*t-1*	−0.079	−0.036	−0.032	0.006	0.002
		(0.050)	(0.038)	(0.031)	(0.026)	(0.024)
	t-2	0.006	−0.021	−0.014	0.028	0.008
		(0.050)	(0.038)	(0.032)	(0.028)	(0.024)
	t-3	0.010	0.0121	−0.014	−0.008	−0.006
		(0.022)	(0.0169)	(0.017)	(0.016)	(0.015)
Financial assets	*t-1*	0.319***	0.239***	0.184***	0.129***	0.109***
		(0.057)	(0.043)	(0.034)	(0.027)	(0.024)
	t-2	0.009	−0.034	0.004	0.069**	0.056**
		(0.060)	(0.046)	(0.036)	(0.031)	(0.024)
	t-3	−0.004	−0.005	−0.024	−0.060**	−0.050**
		(0.041)	(0.032)	(0.029)	(0.025)	(0.022)
Debt	*t-1*	−0.260***	−0.239***	−0.186***	−0.116***	−0.109***
		(0.052)	(0.040)	(0.032)	(0.025)	(0.022)
	t-2	−0.039	−0.048	−0.013	−0.010	−0.007
		(0.053)	(0.043)	(0.035)	(0.027)	(0.024)
	t-3	0.081**	0.088***	0.060**	0.022	0.026
		(0.040)	(0.034)	(0.028)	(0.023)	(0.021)

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TABLE A6 (Continued)

		1971–1990	1976–1995	1981–2000	1986–2005	1991–2010
SV orientation	*t-1*	0.265	−0.144	−1.319***	−1.413***	−1.115***
		(0.462)	(0.323)	(0.262)	(0.230)	(0.164)
	t-2	−0.110	−0.752**	−0.707***	−0.488**	0.155
		(0.401)	(0.336)	(0.252)	(0.216)	(0.164)
	t-3	−0.159	−0.486	−1.481***	−0.610**	−0.666***
		(0.409)	(0.330)	(0.247)	(0.246)	(0.177)
Volatility	*t-1*	−0.070***	−0.093***	−0.082***	−0.046***	−0.033***
		(0.016)	(0.015)	(0.014)	(0.012)	(0.012)
Observations		27,895	34,335	32,933	29,779	27,659
Firms		3,928	4,658	4,656	4,268	3,943
Instruments		1,100	1,800	2,600	3,300	3,100
Firm dummies		Yes	Yes	Yes	Yes	Yes
Year dummies		Yes	Yes	Yes	Yes	Yes
Sargan (*p* value)		0.021	0.020	0.030	0.750	0.985
Second-order auto.		0.469	0.656	0.827	0.418	0.198

Note. Regressions based on the two-step Arellano–Bond difference GMM estimator with Windmeijer (2005) robust standard errors in parentheses. Coefficients for year fixed effects are not reported. (***), (**) and (*) refer to significance at the 1, 5 and 10% level, respectively. For each sub-sample, the instrument set includes lags of the predetermined variables beginning from *t-2*; an identical number of lags for each variable are included in the instrument set, with the instrument set restricted to ensure the number of instruments is less than the number of groups. The size of the instrument set increases for later time periods, due to the existence of additional lags to draw in as instruments. The results are robust to instead restricting each sample to only the sub-period included in the estimation, and imposing an identical instrument set accordingly. The test statistics for the Sargan test of over-identifying restrictions and for the Arellano–Bond test of second-order autocorrelation are reported by their *p* values. Variables are defined in Section 4.4 and summarized in Table A1 in the Appendix.
Source. Compustat, author's calculations.

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